

Human Computer Interaction

Johan F. HOORN

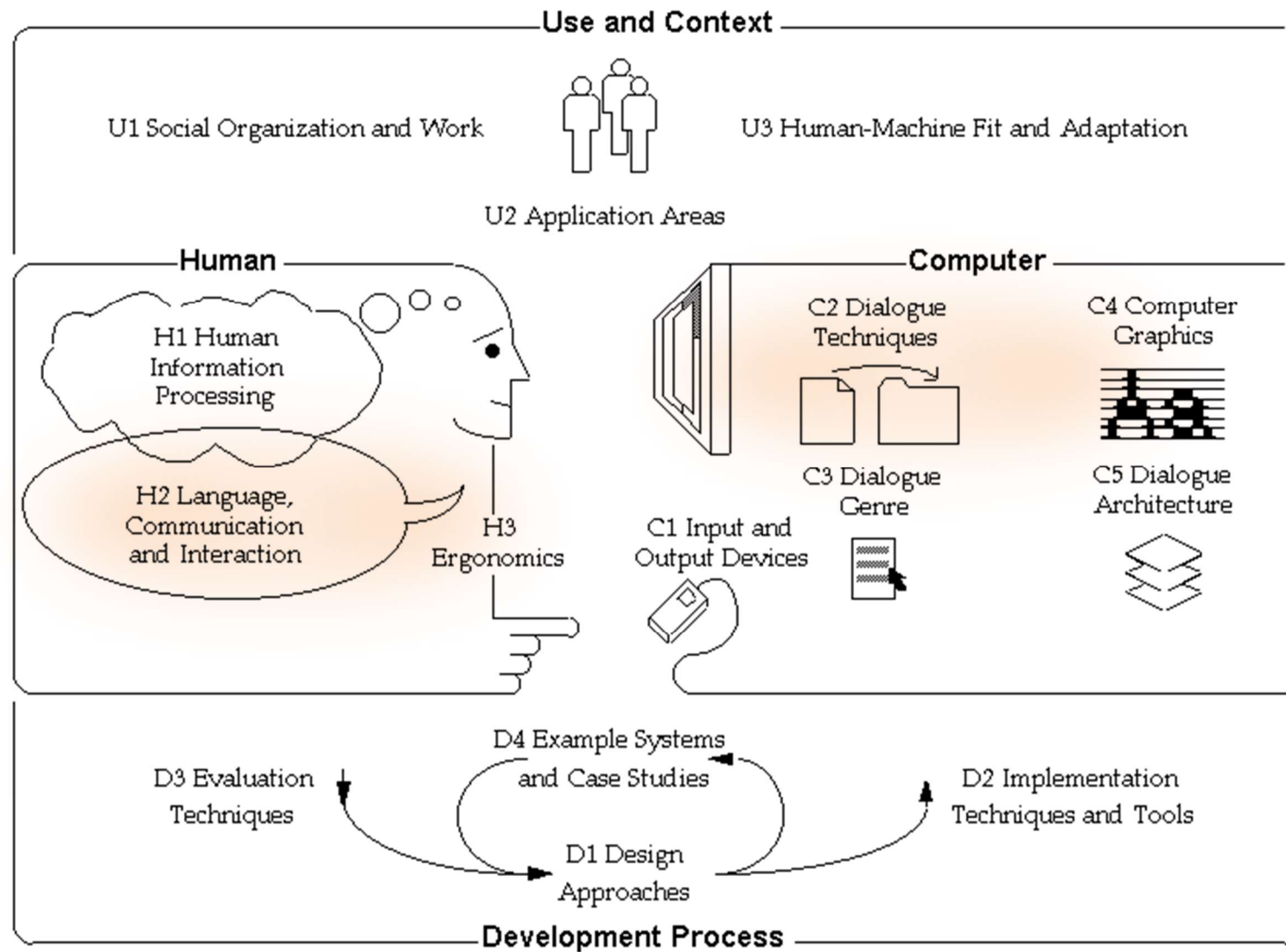
Lecture 04, Part 2: Language, communication, and dialog techniques



C-3PO ("3PO" aka Threepio)

Lukas, G., & Hales, J. (2002). *Star Wars Episode II: Attack of the Clones*. Los Angeles, CA: 20th Century Fox.
[Part 7/10] - 1080p HD, 7:00 min. Retrieved May 21, 2012 from http://www.youtube.com/watch?v=-o5Y_AszZGg

Overview



Contents

- ❑ Lecture's objectives
- ❑ Coding language
- ❑ Dialog languages
- ❑ NLP: Natural Language Processing
- ❑ Human language



Slides in this lecture are partially based on those from Ms. Tiffany Tang, Dr. Vincent Ng, Dr. Grace Ngai, Dr. Saul Greenburg, and Alan Dix et al.

Today:

Machines and humans talking together

Parts of this lecture are from Prof. dr. P. Th. J. M. Vossen and his team
Computational Lexicology, VU University

Why communicate with a machine in natural language?

- Natural language is the most natural and inclusive interface for human interaction
- Flexibility of devices by adaptation through (abstract) language instructions (Mavridis 2015), for example see WikiHow:
<https://www.wikihow.com/Main-Page>
- Non-experts (Mavridis 2015)
 - can use complex devices such as agent systems (cf. Internet of Things)
 - can train and teach devices using natural language
- Many social functions rely on natural language communication, e.g. business transactions
- Systems can operate autonomously (after briefing) and report back to you about experiences
- Language is the most efficient way of communication: abstract, minimalistic, and exploiting common ground

Machines and humans talking together

Machines and humans talking together



Kubrick, S. (1968). 2001 A Space Odyssey

Machines and humans talking together

❑ Natural Language Processing (NLP)

Natural Language

- A language that is spoken, written or signed by humans for general communication purposes
- Contrasted with formal languages (programming, mathematical languages) and constructed languages
- Used heavily in HCI interfaces in science fiction!



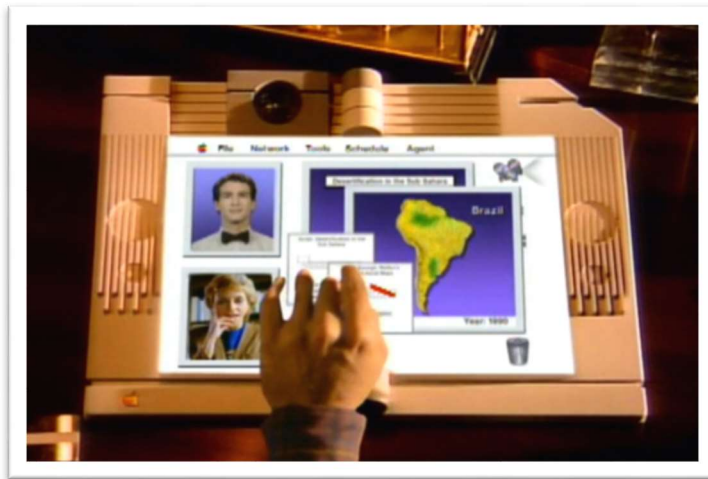
Lukas, G., & Hales, J. (2002). *Star Wars Episode II: Attack of the Clones*. Los Angeles, CA: 20th Century Fox.

Machines and humans talking together

□ Natural Language Processing (NLP)

Natural Language

- Used heavily in HCI interfaces in science fiction!



Apple's Knowledge Navigator (1987):
<http://www.youtube.com/watch?v=HGYFEI6uLy0>



'Her' the movie (2013):
<https://www.youtube.com/watch?v=WzV6mXIOV14>
[http://en.wikipedia.org/wiki/Her_\(film\)](http://en.wikipedia.org/wiki/Her_(film))

Machines and humans talking together

□ Natural Language Processing (NLP)

Natural Language

- Not just science fiction!
 - Deep Blue, Alpha Go, Deep Mind
 - Apple's Siri, Google Home, Amazon Alexa, Facebook Messenger Bot, Microsoft Cortana
 - IBM Watson winning Jeopardy!
<https://www.youtube.com/watch?v=yKN8a8jgIN8>
 - Google Duplex making appointments
<https://youtu.be/D5VN56jQMWM>
 - IBM Debater
<https://www.research.ibm.com/artificial-intelligence/project-debater/>
<https://www.youtube.com/watch?v=naQujxmg9gg>
 - GPT4 and beyond



Machines and humans talking together

□ Natural Language Processing (NLP)

Difficult: variation in meaning (e.g., synonyms)

How many words for a person? Man, guy, bloke, gentleman, lad, ...



Which English noun describes this man best?
The Joker is a ...

English has >5000 words for a person

mortal; posturer; controller; **withholder**; suppressor; subduer; fugitive; divider;
subdivider; outcaste; bereaved_person; yielder; nude; streaker; **unperson**;
baby_buster; neighbor; loose_cannon; ladino; communicator; announcer; town_crier;
caller; caller; muezzin; hisser; gossipmonger; yenta; cat; telltale; **scandalmonger**;
allegoriser; presenter; promisee; quoter;

transmitter; spammer; answerer; hedger; assenter; interviewee; **don'tknow**; testee;
passer; popularizer; avower; laudator; clapper; waffler; wirer; conferrer; confessor;
reporter; newswoman; television_reporter; anchorman; avower; postulator; author;
gagwriter; poet; sonneteer; poetess; homer; elegist; frost; key; gilbert; gray; pound;
poet_laureate; odist; spender; poet_laureate; bard; biographer; autobiographer;
hagiographer; novelist; folk_writer;

folk_poet; **cyberpunk**; west; wood; authoress; abstractor; pamphleteer; speechwriter;
drafter; paragrapher; space_writer; tragedian; snow; day; playwright; rice; kid; cooper;
buck; scriptwriter; film_writer; wordmonger; framer; literary_hack; rand; coauthor;
scenarist; litterateur; **word-painter**; wordsmith; alliterator; sand; e._e._cummings;

English has 4,000 words for move

go; swim; buoy; drive; island_hop; whistle; ski; slalom; hot-dog; wedel; water_ski; schuss; pass_over; breeze; err; return; revisit; retrace; cut_back; resurrect; return; home; head_home; double_back; bounce; boomerang; fly; come; retrograde; walk; speed; bang; **swash**; tread; step_on; beetle; circulate; drift; swim; bucket_along; shoot; rip; barge; dash; plunge; hurtle; sit; canter; override; prance; ride_herd; ride_horseback; **prance**; post; trot; gallop; canter; gallop; outride; lance; scramble; plough; sift; **zigzag**; billow; pursue; haunt; tail; tree; hound; ferret; run_down; quest; stalk; roll; troll; bowl; travel_purposefully; wend; whisk; cruise; stooge; steamer; go_forward; head; make; trace; limp; wander; roar; ease; circulate; float; ride; shack; draw; caravan; career; raft; swap; thrash; retreat; cocoon; automobile; step; backpedal; blow; stream; tide; waft; crawl; fornicate; slice_into; run; precede; lead; draw_away; travel_along; ascend; heel; turn; angle; push; travel; ride; fly; cruise; ship; sail; wind; snake; pan; repair; taxi; precess; cast; **jazz_around**; maunder; travel; itinerate; go_up; rise_up; resurface; well; intumesce; emerge; uprising; ferry; transfer; betake_oneself; march_on; string; edge;

English has 1,148 words for sound

roll; vibrate; resonate; make_vibrant_sounds; purr; gurgle; guggle; ripple;
bubble; babble; burble; beat; drum; thrum; squelch; guggle; crash; swoosh;
swish; lap; swosh; trump; rattle; ruckle; crackle; crepitate; birr; whir; whiz; whirr;
whizz; purr; skirl; blow; rustle; ticktack; beat; ticktock; tick; twang; boom_out;
boom; bleep; whistle; sing; ring; peal; **knell**; dong; ding; dingdong; peal;
tintinnabulate; gurgle; slosh; slush; splosh; splash; pitter-patter; patter; bang;
crack; thud; thump; grumble; rumble; blow; splat; clang;
clangor; whish; beep; toot; blare; claxon; honk; tootle; snarl; hum; thrum;
boom; din; ting; resound; make_noise; noise; blast; blare; scream; screech;
creak; whine; squeak; skreak; claxon; honk; scraunch; crackle; crunch; scrunch;
crump; thud; scrunch; drown_out; jingle- jangle; jangle; jingle;
stridulate; clitter; ring_out; scream; sizzle; sough; purl;
racket; backfire; hum; roar; howl; thunder; vroom; bawl; yawp; clack; clatter;
brattle; chug; **glug**; tinkle; clink; tink; chink; rap; tap; knock; pink; clop; plunk;
clump; clunk; resound; echo; ring; reverberate; bong; consonate; **reecho**; click;
tick; drone; crack; snap; bombilate; buzz; bombinate; twirp; tweet; chime; clank;
ping; knock; pink; ping; clangor; clangour; chatter; click; whistle; pop; sputter

English has 2,037 words for noise

grinding; racket; report; squeak; clap; clack; snore; chatter; chattering; brouhaha; hubbub; uproar; **katzenjammer**; clatter; shrieking; scream; screech; screaming; shriek; screeching; blaring; blare; din; cacophony; clamor; grumble; rumble; grumbling; rumbling; squawk; plump; crackling; crackle; crepitation; decrepitation; snap; explosion; squish; rhonchus; hum; humming; **swoosh**; **whoosh**; clangoring; clang; clank; clash; clangor; crash; clangour; whisper; whispering; rustling; rustle; chug; sizzle; plonk; howl; **squeal**; plop; scrape; scratching; scratch; scraping; chatter; chattering; ding-dong

Few words for smell/taste

smell; salute; reek; get_a_noseful; wind; snuffle; sniff; smelling; snuff; odor; fragrance; rankness; body_odor; rancidness; muskiness; olfaction; nose; olfactory_sensation; odor; olfactory_perception; smell; odour; fragrance; incense; acidity; scent; musk; malodor; niff

taste; salinity; sugariness; astringency; savour; lemon; vanilla; finish; sourness; acidulousness; bitter; mellowness; flatness

BUT people can distinguish more than 1 trillion different smells, several millions of colors and half a million tones: [Humans can discriminate more than 1 trillion olfactory stimuli](#)

Bushdid C, Magnasco MO, Vosshall LB, Keller A. Science. 2014 Mar 21;343(6177):1370-2. doi: 10.1126/science.1249168.

Ambiguity is pervasive and we do not perceive it!!!

- 121 most frequent English nouns have on average 7.8 meanings each and account for about 20% of word occurrences in real text (in the Princeton WordNet (Miller, 1990), according to Ng and Lee (1996))
- “He gave a soft ball across the line from the center of the field, making a major point and giving a minor lead”

Ambiguity demo

Sentence:

Compute

He(2) gave(44) a soft(19) ball(12) across the line(30) from the center(18) of the field(17) , making(49) a major(8) point(26) and giving(44) a minor(10) lead(17) . = 14041749244723200 possible meaning combinations

14,041,749,244,723,200 = 14 quadrillion combinations !!!!

Computer -- There's
a power reduction in
section L-94. Explain.

40271
ACCESS=11001001
FOUND

Unoccupied sections are being closed down in order to clear space
in the main computer banks. It's necessary to facilitate the
system upgrade. Full power can be restored if needed.

1988 Star Trek: The Next Generation: Season One — Episode Title: 11001001

A man with dark hair, seen from the side, wearing a red Star Trek uniform, is looking at a computer screen. The screen displays a dark interface with orange and blue text and graphics.

40271

ACCESS=11001001

FOUND

What would be needed for dialog like this to work?

Machines and humans talking together

❑ Natural Language Processing (NLP)

Understanding humans



There's a power reduction in section L-94. Explain.

phenomenon

location

request

Discourse Analysis

- “There's a power reduction in section L-94. Explain (it).”
- What does the “it” refer to?
- *Coreference resolution* links together noun phrases (pronouns, names, noun groups) into entities (things that actually exist).
- “It” here is linked to “power reduction”
- Usually done via *heuristics*
- Looks at gender, number, arguments of phrases, etc.

Coreference resolution is the task of finding all expressions that refer to the same entity in a text. It is an important step for many higher level NLP tasks

Machines and humans talking together

□ Natural Language Processing (NLP)

Semantic Analysis

- Marks up text with semantic (meaning) information, such as proper names, gender, number, cause, effect, etc.
- Uses heuristics combined with statistics and probabilities

gender: neutral
number: singular
type: location
animate: no

There's a power reduction in section L-94. Explain.

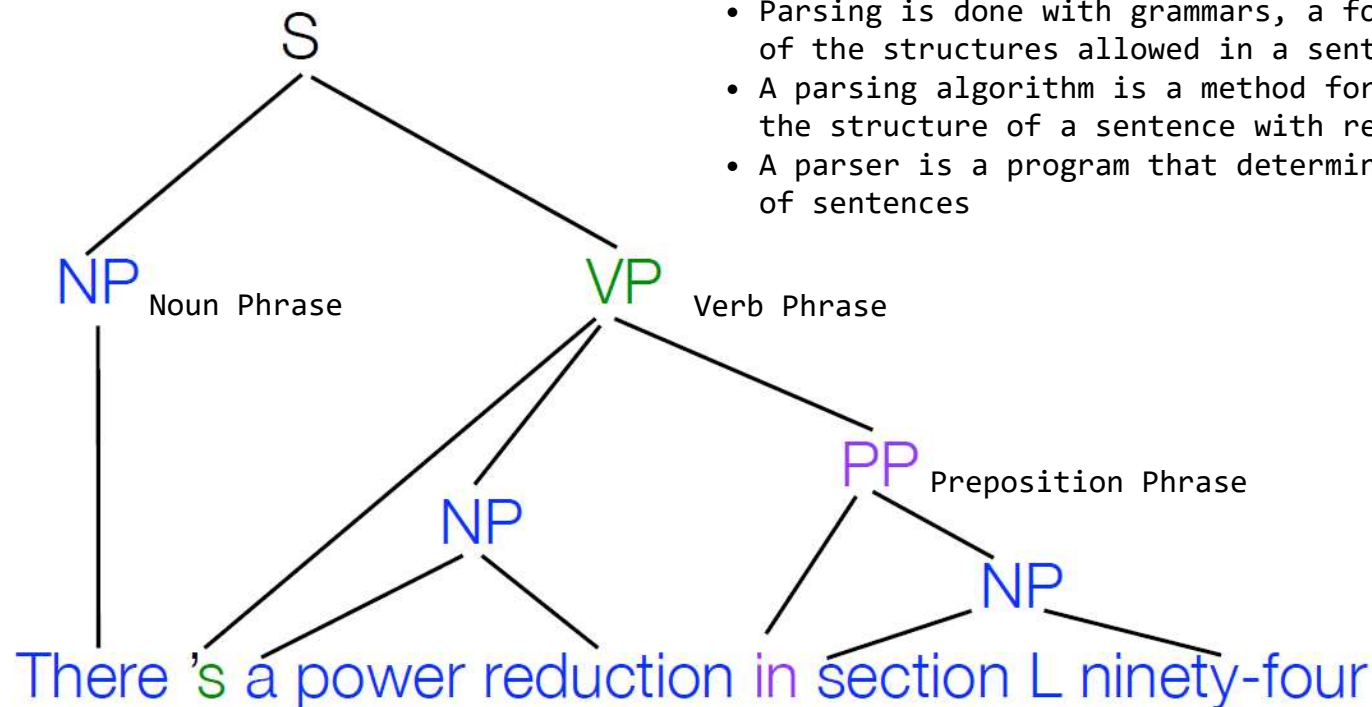
gender: neutral	proper
number: singular	name
type: abstract	
animate: no	

Machines and humans talking together

□ Natural Language Processing (NLP)

Syntactic Analysis (Parsing)

- Analyzes groups of related tokens (words) that form larger constructs (sentences)
- Converts linear sentence into hierarchical structured parse tree



Machines and humans talking together

□ Natural Language Processing (NLP)

Part-of-speech tagging

- Part-of-speech tagging is the task of labeling each word in a sentence with its appropriate part-of-speech tag
- POS classes group words into different classes depending on their syntactic or grammatical behavior

Classes include:

- Nouns (people, animals, concepts and things)
- Verbs (actions)
- Adjectives (properties of nouns)
- Adverbs (properties of verbs)
- And many other classes, such as pronouns, prepositions, determiners, etc.

There/DT 's/VB a/Art power/NN reduction/NN in/Prep section/NN
L/NN ninety-four/NUM

Machines and humans talking together

□ Natural Language Processing (NLP)

Probabilistic POS tagging

- A word can have many tags!
E.g., “work” can be a noun, or a verb

Ambiguity demo

Sentence:

He(2) gave(44) a soft(19) ball(12) across the line(30) from the center(18) of the field(17) , making(49) a maj

Compute

He(2) gave(44) a soft(19) ball(12) across the line(30) from the center(18) of the field(17) ,
making(49) a major(8) point(26) and giving(44) a minor(10) lead(17) . = 14041749244723200
possible meaning combinations

Machines and humans talking together

□ Natural Language Processing (NLP)

Probabilistic POS tagging

- A word can have many tags!
E.g., “work” can be a noun, or a verb
- Usually this is done using probabilities:
 - Simple probability: $t_i = \operatorname{argmax}_t P(w_i, t)$

Argmax is a mathematical operation, looking for the argument that yields the highest value (the max) from a target function. Finds the candidate with the largest expected probability

Machines and humans talking together

□ Natural Language Processing (NLP)

Probabilistic POS tagging

- Usually this is done using probabilities:
 - Simple probability: $t_i = \operatorname{argmax}_t P(w_i, t)$
 - However, we would like to add in contextual information.
Given w , what is the most likely t ?

$$t_{1,n} = \operatorname{argmax}_{t_{1,n}} \prod_{i=1}^n [P(w_i | t_i) \times P(t_i | t_{i-1})]$$

Π = multiplication
 $P(t_1 | t_0) = 1$

Now it searches for the highest P , the most likely tag sequence ($T = t_1, \dots, t_n$),
given a certain word sequence ($W = w_1, \dots, w_n$)

- Yet, ineffective if related words are separated in the text
- Looks back, does not predict context/words about to come
- GPT solved many of these issues

Natural Language Processing (NLP)

Generally, NLP models try to predict what word will follow in a sentence (next-token-prediction). Tokenization is identifying words, numbers, operators, etc.:

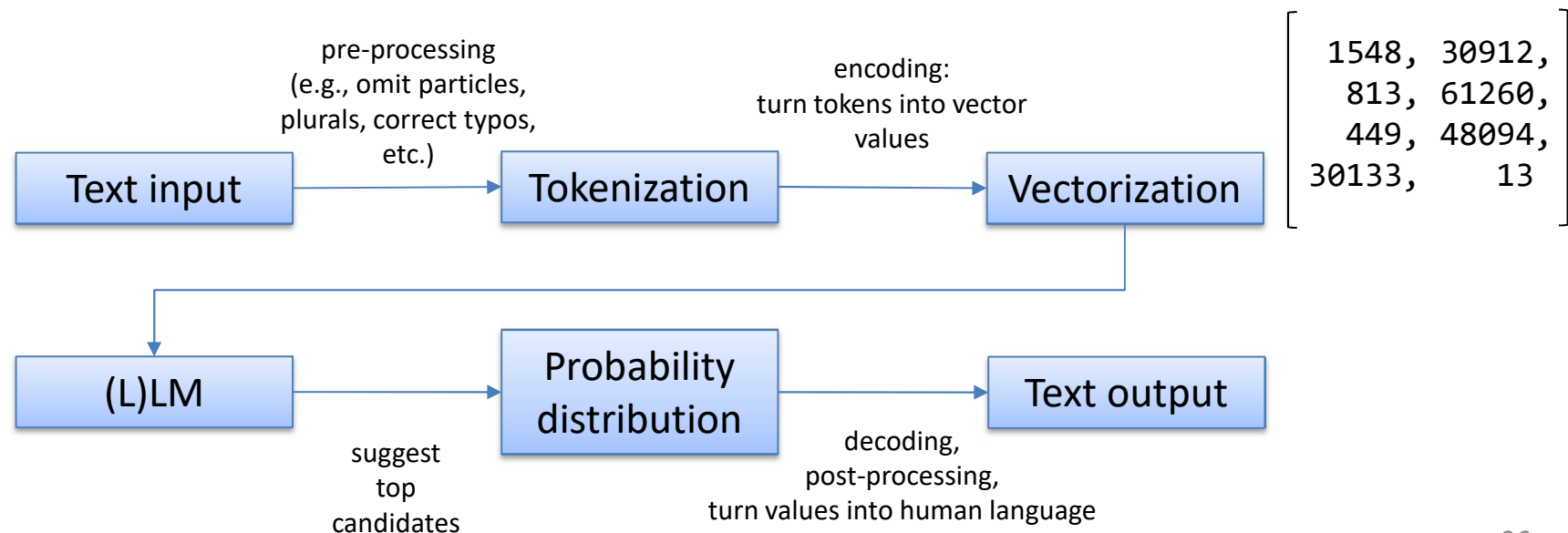
“He ate his noodles with ...”

‘motor oil’ has low likelihood

Predicting a missing word like in the example is called ‘masked language modeling’

The word with the highest probability is not always the best choice. Therefore, some randomness is added (or ‘temperature’) to be able to vary the wording

General NLP approach:



Machines and humans talking together

❑ Natural Language Processing (NLP)

Tokenization

- Converts strings of characters into words
- After taking a sample from a dataset, we tokenize and change it into a vector
- This just means putting numbers to words, which fits computer processing better

Input Sentence	Encoding	Tokenization & Vectorization	Internal Representation
He ate his noodles with chopstick.	English	[1548, 30912, 813, 61260, 449, 48094, 30133, 13]	['He', ' ate', ' his', ' noodles', ' with', ' chop', 'stick', '.']
他用筷子吃麵。	Traditional Chinese	[43511, 11883, 29857, 115, 45829, 7305, 225, 165, 118, 113, 1811]	'他用筷子吃麵。' (notice this isn't cut. There is ongoing research)
他用筷子吃面条。	Simplified Chinese	[43511, 11883, 29857, 115, 45829, 7305, 225, 28190, 40089, 1811]	'他用筷子吃面条。' (notice this isn't cut. There is ongoing research)

Machines and humans talking together

□ Natural Language Processing (NLP)

Tokenization

- Trivial (almost) for Indo-European languages
- More complex for other languages, e.g.:
 - Chinese needs word segmentation. Why is Chinese hard?

There's a power reduction in section L-94. Explain.

新华社上海二月十日电（记者谢金虎、张持坚）上海浦东近年来颁布实行了涉及经济、贸易、建设、规划、科技、文教等领域的七十一件法规性文件，确保了浦东开发的有序进行。

Word segmentation gives:

新华社上海二月十日电（记者谢金虎、张持坚）上海浦东近年来颁布实行了涉及经济、贸易、建设、规划、科技、文教等领域的七十一件法规性文件，确保了浦东开发的有序进行。

Machines and humans talking together

□ Natural Language Processing (NLP)

Tokenization

- Turkish needs intra-word tokenization. Why is Turkish hard?

- jazusilarimizdayilardan
- jaz: write
- jaz.u: letter, writing
- jaz.u.si: writer
- jaz.u.si.lar: writers
- jaz.u.si.lar.im: my writers
- jaz.u.si.lar.im.iz: our writers
- jaz.u.si.lar.im.jz.da: appertaining to our writers
- jaz.u.si.lar.im.iz.da.yi: that which our writers have
- jaz.u.si.lar.im.iz.da.yi.lar: those which our writers have
- jaz.u.si.lar.im.iz.da.yi.lar.dan: from those which our writers have

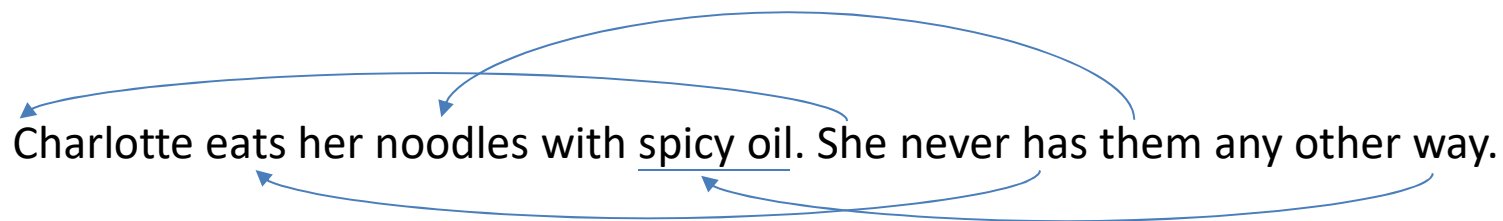
GPT (Generative Pre-trained Transformer) by *OpenAI*

- Generative: It generates output
- Pre-trained: Uses large corpora of text with human feedback



- What is a Transformer?

A neural-network architecture (see next slides), trying to connect tokens to one another based on a similarity measure



Somewhat overstated, this similarity estimate is called an 'attention' mechanism (see Vaswani et al., 2017)

Inside GPT's Transformer

Positional Encoding

Positional encoding defines the location of a token in a sentence or other sequence and gives that position a unique value

<i>Sentence</i>		<i>Index of the tokens (= words)</i>		<i>Matrix in which the position of the token in the sentence is encoded</i>			
I	→	0	→	P_{00}	P_{01}	...	P_{0n}
am	→	1	→	P_{10}	P_{11}	...	P_{1n}
a	→	2	→	P_{20}	P_{21}	...	P_{2n}
robot	→	3	→	P_{30}	P_{31}	...	P_{3n}

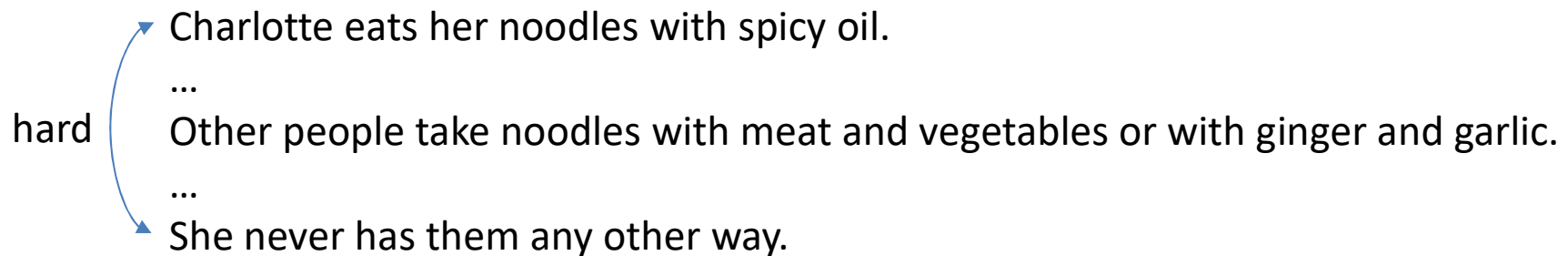
Now different attention mechanisms can operate on that encoded piece of text

Inside GPT's Transformer

After Positional Encoding: Attention modules

- Regular attention (early approach): Connects an input to an output sequence (A co-occurs with B)

The further away words are, the harder it becomes to link them together. So proximity is not good to account for semantic association



Inside GPT's Transformer

Attention modules

- Self-attention (provides some context): Focuses on a single sequence / sentence. Links different positions of that single sentence together and calculates a representation about that sequence. The sentence learns about itself

- The system sees a word as a token
- Every token has a word embedding, without context
- To provide context to the word embedding, weights are added based on similarity

For instance, the words 'noodles' and 'eat' *probably* belong together no matter where they are in the sentence. Words like 'and' and 'or' are less related to Charlotte than the word 'she'

Inside GPT's Transformer

Attention modules that GPT uses

- Single-head attention looks at one aspect of the sentence, then at another
- Multi-head attention: Each analysis or 'head' runs in parallel, paying attention to different aspects of the sentence. For instance, Head 1 guesses the next word (motor oil?) while Head 2 analyzes how adjectives and nouns should be put together (spicy oil, not oil spicy). Multi-head attention is a parallel process, estimating similarity among tokens that are then concatenated (weighted sum) for the most probable output: "She eats noodles with spicy_oil"

Head 1 or Analysis 1

... eats noodles with _____

spicy oil	$w = 80\%$
garlic oil	$w = 10\%$
motor oil	$w = 3\%$
etc	

Head 2 or Analysis 2

... eats noodles with _____

spicy + oil	$w = 40\%$
oil + spicy	$w = 15\%$
motor + oil	$w = 40\%$
etc	

Inside GPT's Transformer

Attention modules that GPT uses

- Single-head attention looks at one aspect of the sentence, then at another
- Multi-head attention: Each analysis or 'head' runs in parallel, paying attention to different aspects of the sentence. For instance, Head 1 guesses the next word (motor oil?) while Head 2 analyzes how adjectives and nouns should be put together (spicy oil, not oil spicy). Multi-head attention is a parallel process, estimating similarity among tokens that are then concatenated (weighted sum) for the most probable output: "She eats noodles with spicy_oil"

Head 1 or Analysis 1

... eats noodles with _____

spicy oil	$w = 80\%$
garlic oil	$w = 10\%$
motor oil	$w = 3\%$
etc	

Head 2 or Analysis 2

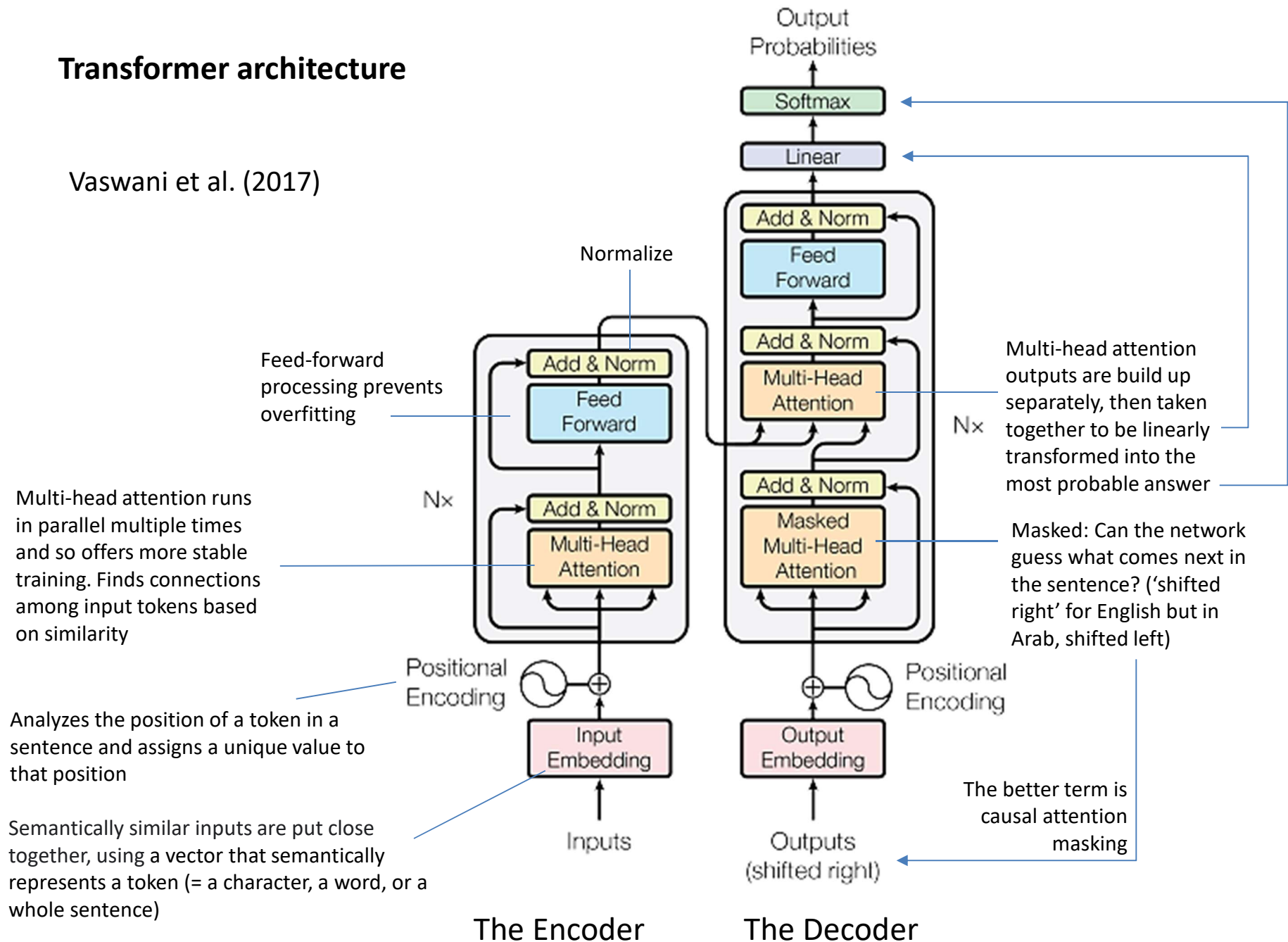
... eats noodles with _____

spicy + oil	$w = 40\%$
oil + spicy	$w = 15\%$
motor + oil	$w = 40\%$
etc	

Best candidate
is selected

Transformer architecture

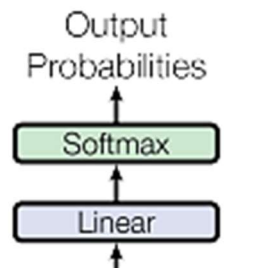
Vaswani et al. (2017)



GELU (Gaussian Error Linear Units) activation function

In its transformer network, GPT also uses a special activation function

The GELU function is an approximation of the ReLU (Rectified Linear Unit) activation function. Provides computational efficiency in DL neural networks. At the end of its process (the output layer), GPT applies the Softmax function to simplify the results



Next, we will attempt to understand GELU, what it avoids compared to ReLU, and what Softmax is doing

But before that, let's look at activation functions again. What are they?

Activation functions in general

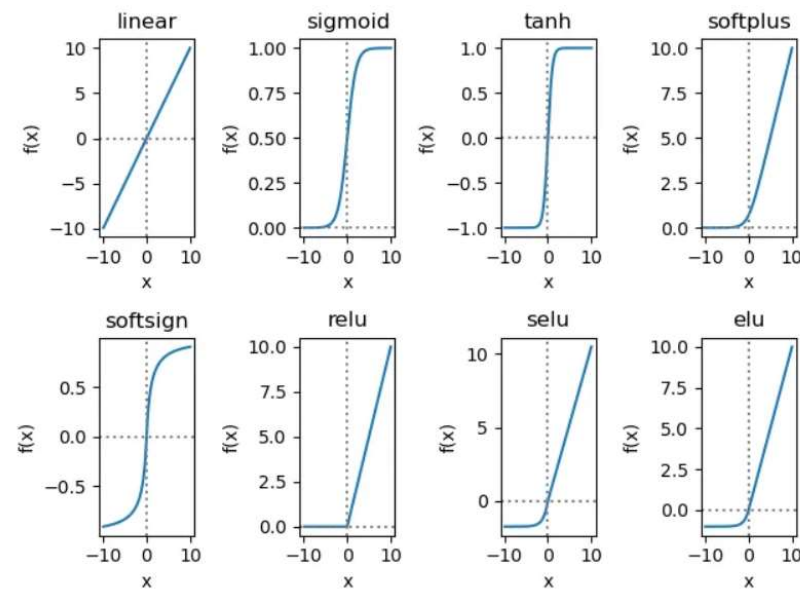
slope bias

In principle, a NN is linear ($ax + b$). Adding an activation function makes the NN non-linear and that allows the neural network to 'learn' complex patterns and make predictions

Non-linearity would better fit most data sets and improve accuracy (hopefully)

An activation function calculates the weighted sum while adding bias

When predictions are wrong, back propagation returns the error to the hidden layers so to adjust the weights

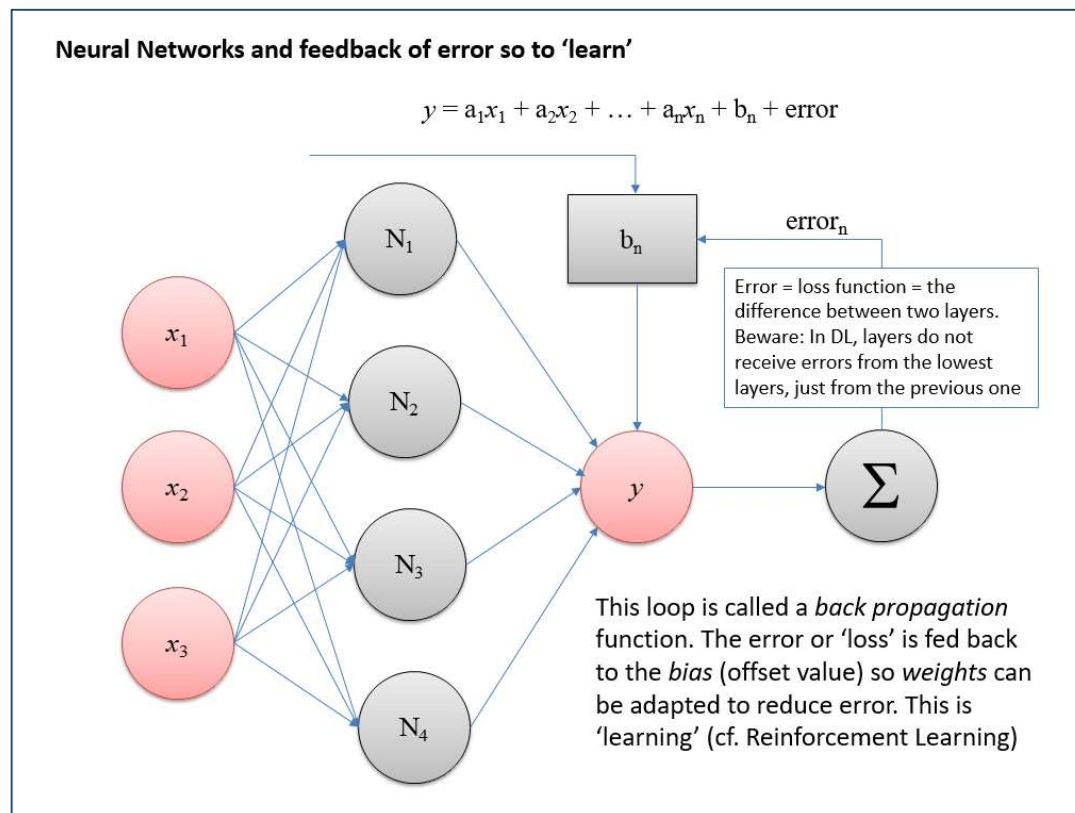


What is the 'vanishing gradient' problem?

We have weights connected to words (x) and the weights tell the next layer of the network how much the output (y) is affected by that word

Those weights are also called gradients

Gradients can be adjusted to minimize error (= maximize model performance)



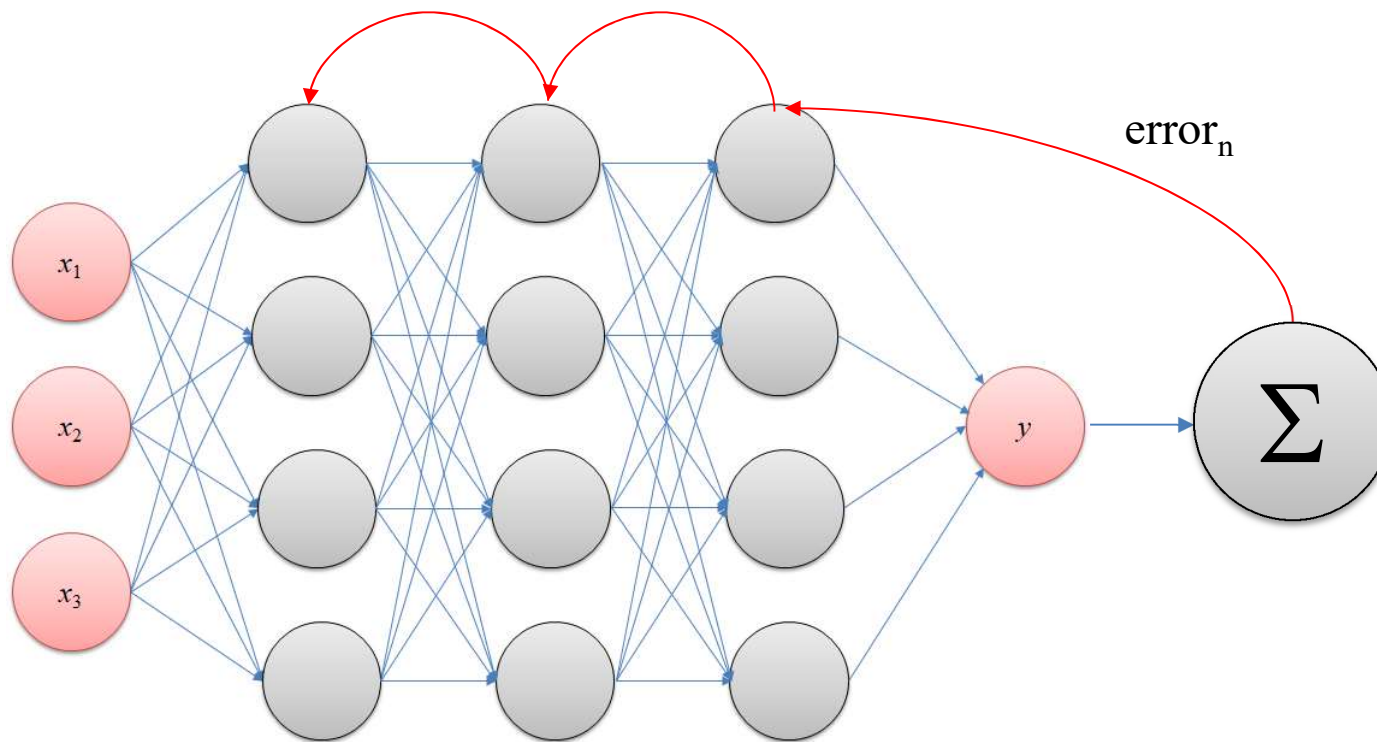
What is the 'vanishing gradient' problem?

Backpropagation calculates the gradients of the error (loss) for each weight

It does that one layer at a time, going backward from the layer closest to the output (y)

... to the layer closest to the input (x)

Feed back error step by step from the last layer to the first

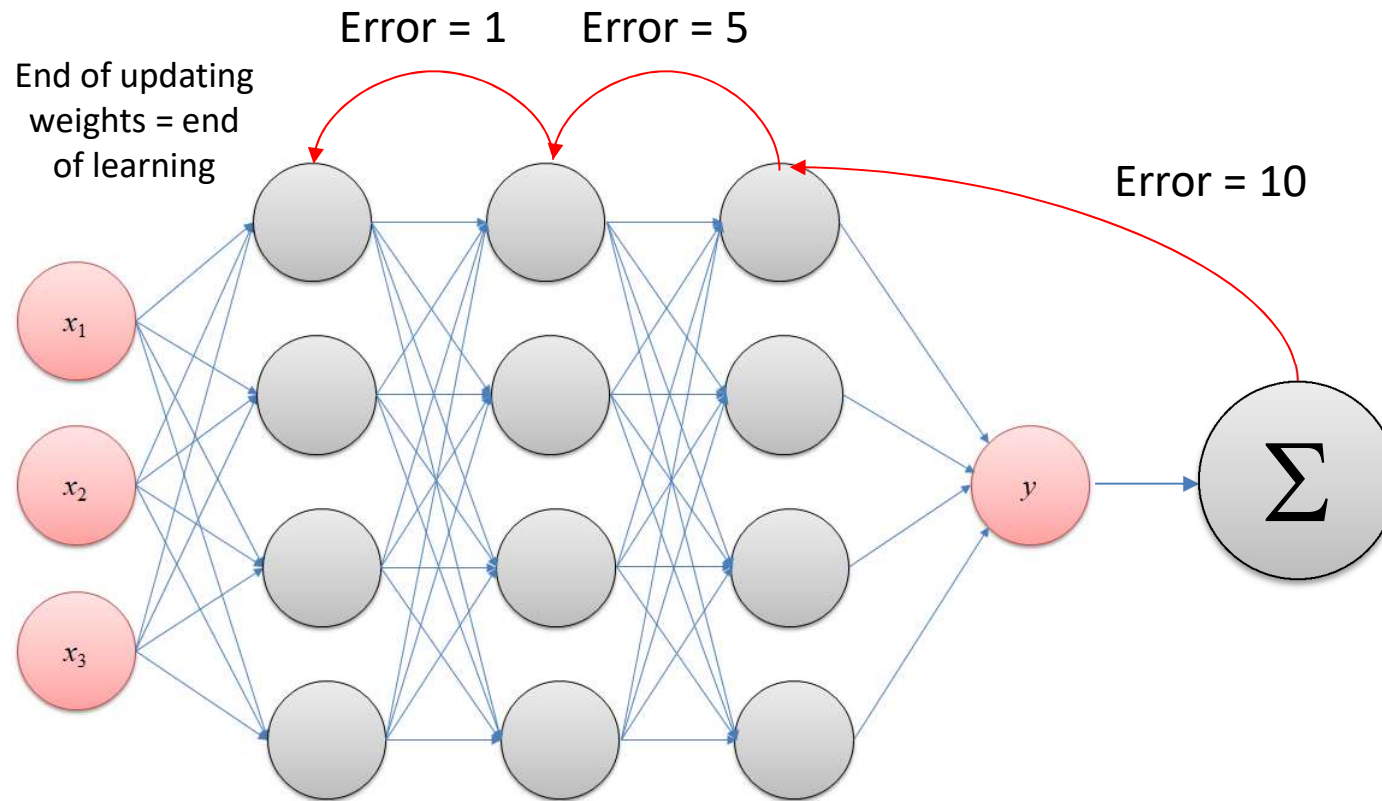


What is the 'vanishing gradient' problem?

Gradients become smaller and smaller with each layer we go back. They 'vanish'

Like that, the weights in the first layers (close to x) are hardly updated anymore

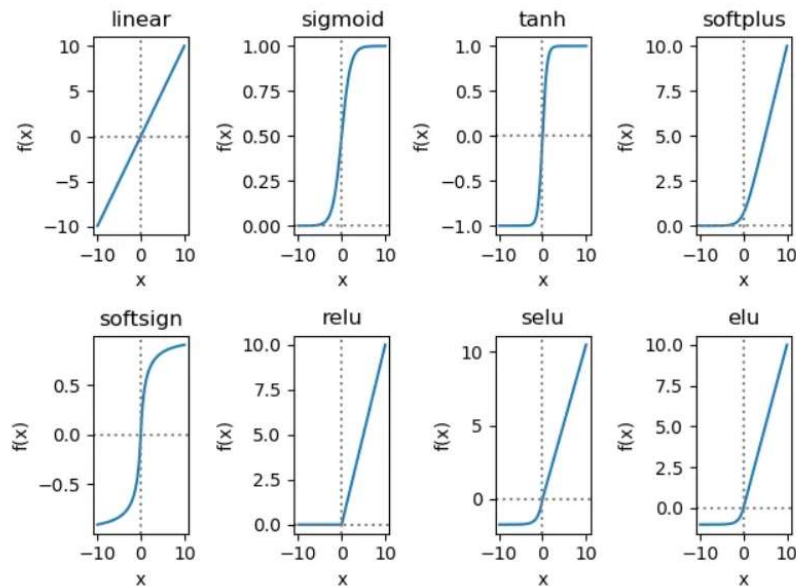
Thus: no learning!!



What causes the 'vanishing gradient' problem?

Activation functions!!!! Particularly the older ones

For example, sigmoid or hyperbolic tangent (tanh) compress input into a small range (sigmoid: 0 to 1; tanh: -1 to 1). Through the layers, gradients become smaller and sometimes disappear

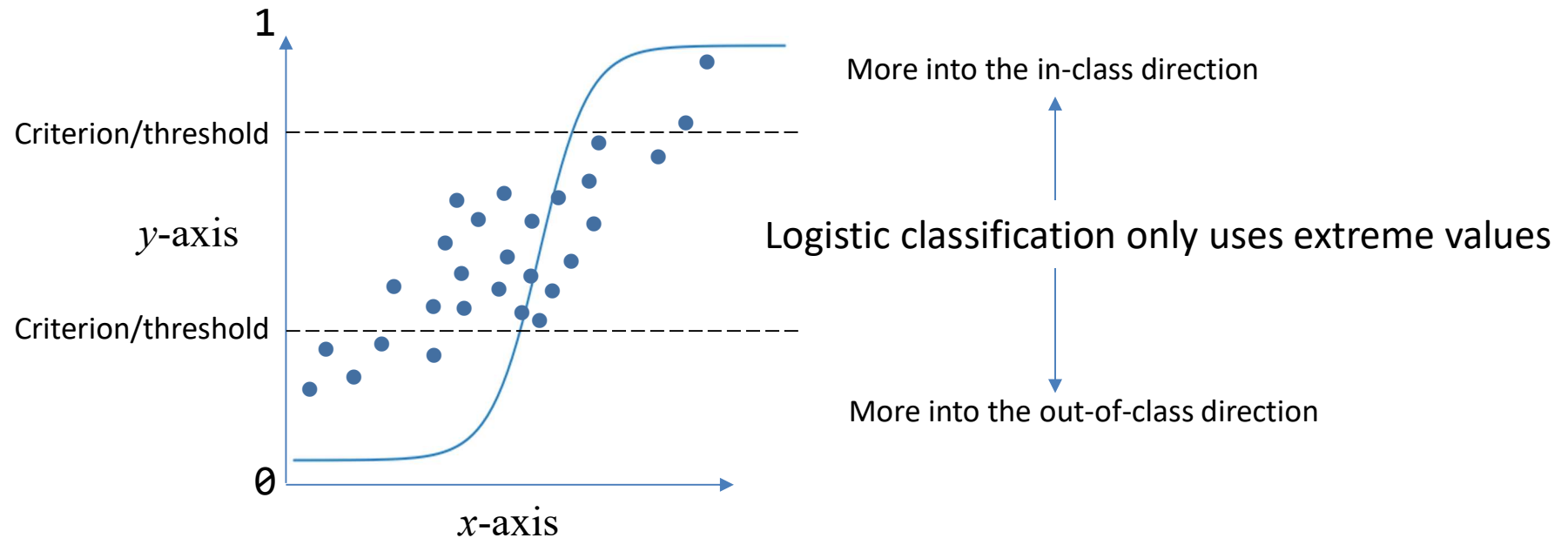


Sigmoid (logistic regression) and tanh suffer from 'vanishing gradients,' which are solved by using ReLU or as GPT does by using the GELU activation function

Note: Residual networks are another option but that is for the advanced course 😊

The vanishing gradient problem

Logistic regression (sigmoid function) pushes data either in or out of class



Thus, a large data set becomes a small data set with values between 0 and 1 (the 'derivative of the activation function')

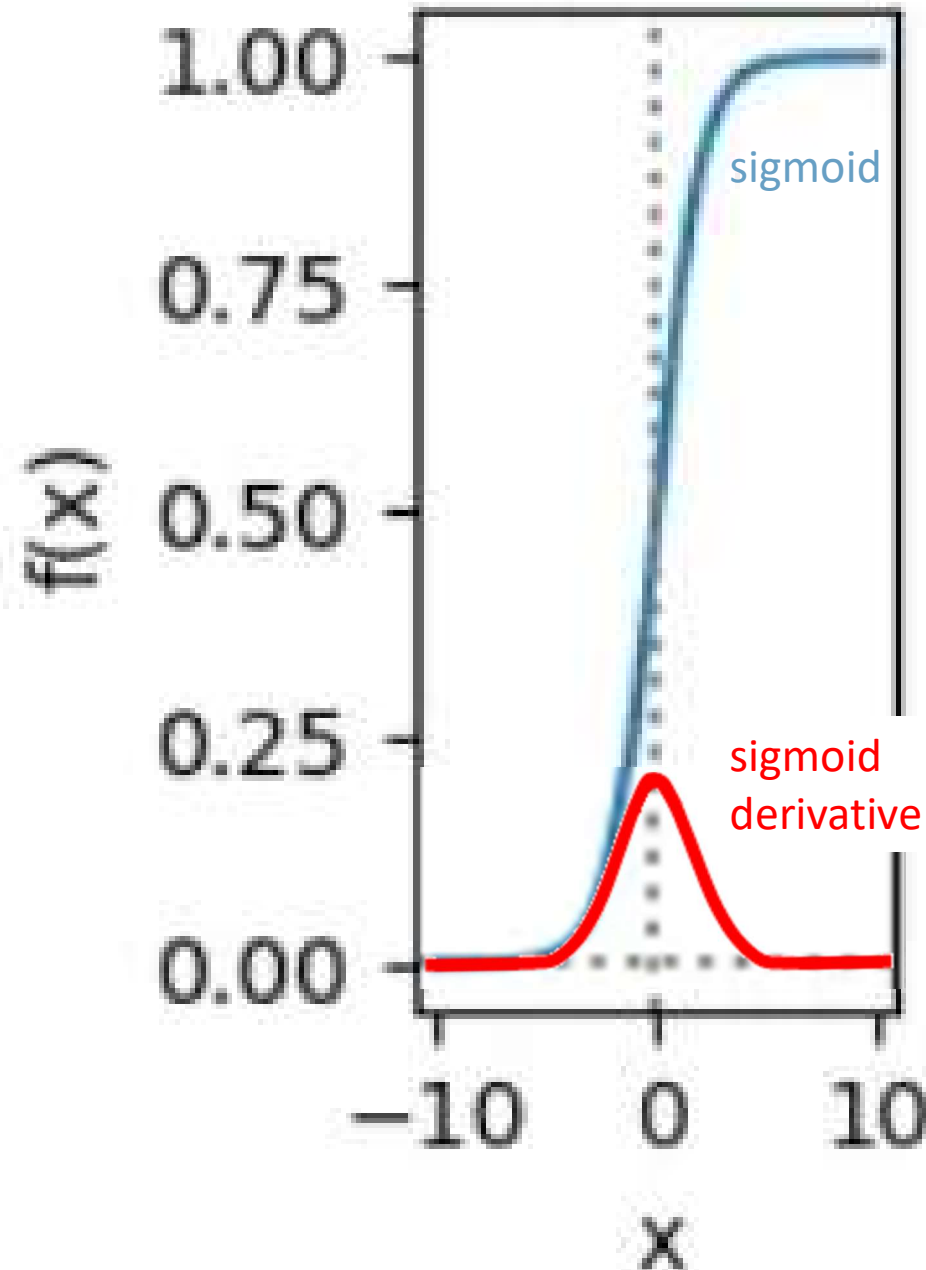
Therefore, the sigmoid turns big differences between input values (e.g., 1 and 100) into small differences (e.g., .1 and .9). This is called the 'derivative' of the activation function and *sigmoid* and *tanh* make the derivative small

Overcoming the vanishing gradient problem

Observe the **derivative** of the sigmoid: with bigger amounts of input, x becomes larger, such that the derivative (that what the sigmoid function outputs) approaches 0

Not a big deal for a single NN but if you insert more and more layers into your DL, you add more activation functions and the gradients of the loss function (= error) go to 0, which hampers training the network

ReLU does not render small derivatives



Rectified Linear Unit (ReLU)

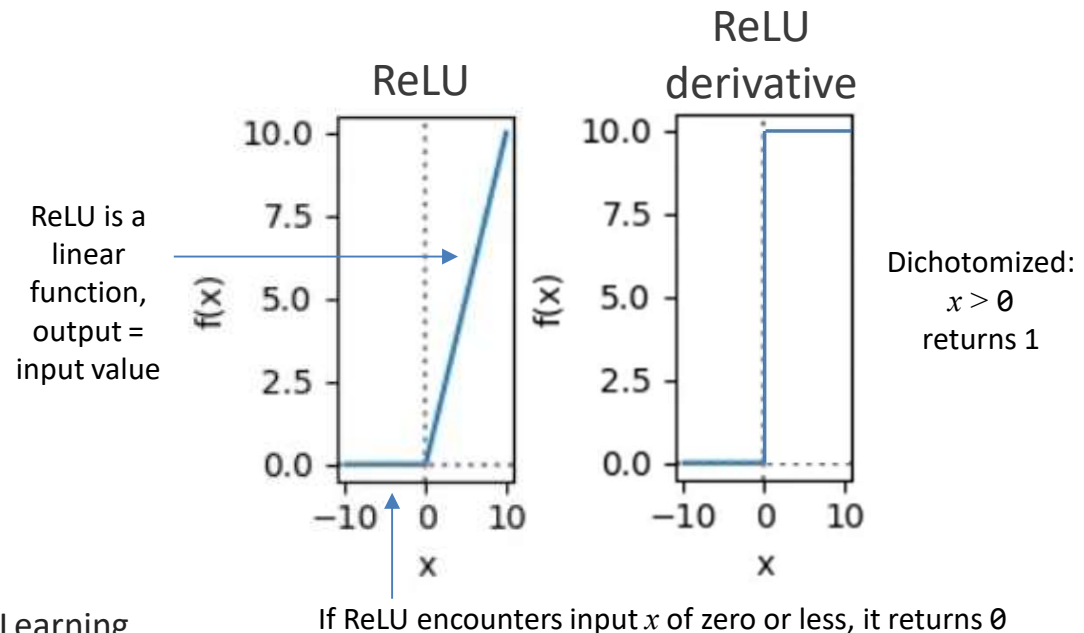
ReLU takes care of the vanishing gradient problem (see the derivative)

ReLU: $f(x) = \max(0, x)$
(output range between 0 and ∞)

- Makes NN non-linear
- Removes negative inputs (x below 0)
- Mainly in hidden layers
Softmax in last layer, see next slide
- Does not activate all neurons at once
- Quick convergence:

If you make a NN larger (a very Deep Learning system), computational cost may increase exponentially. Yet, putting in more ReLU algorithms increases computation only linearly (making it faster)

- Therefore, NN may quickly overfit
- ReLU variants: SWISH and MISH often seen as 'better' but GPT uses GELU
<https://krutikabapat.github.io/Swish-Vs-Mish-Latest-Activation-Functions/>



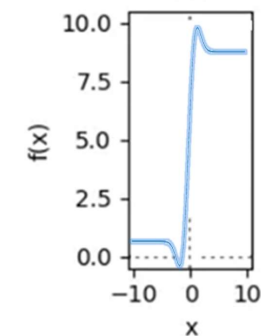
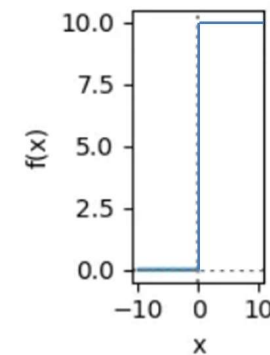
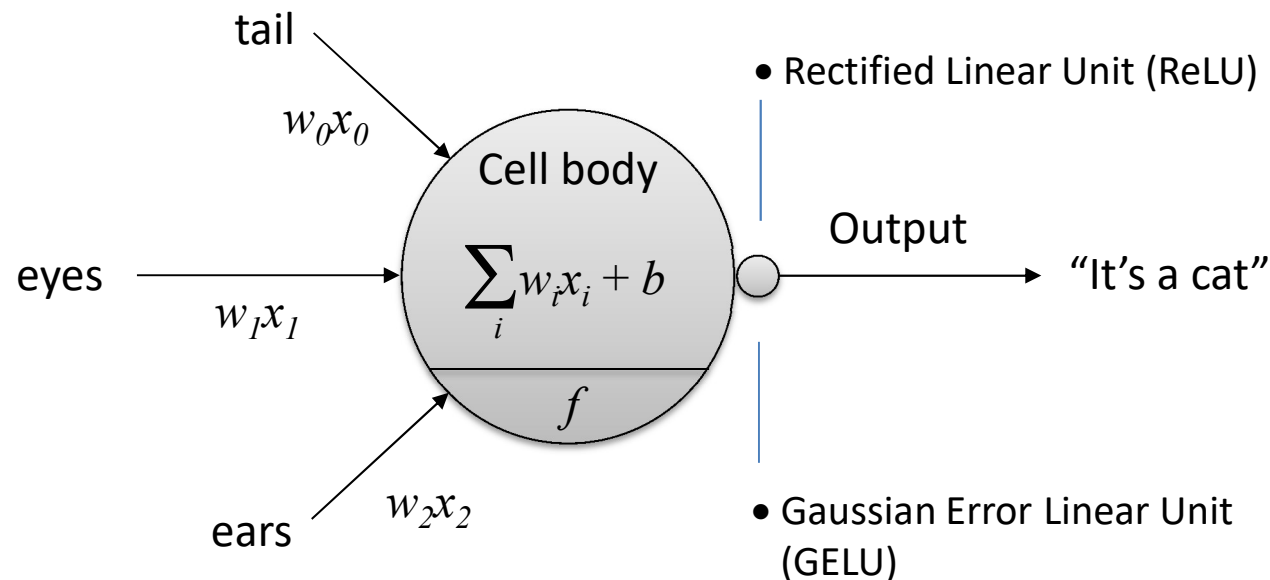
Gaussian Error Linear Unit (GELU)

GELU is what GPT uses so error can be propagated back without vanishing gradients (weights are updated also in the early layers of the deep network)

Most older activation functions, ReLU included, are not curvi-linear and handle complex data sets less well

GELU is more stochastic, is more probabilistic, and therefore can handle complexity better

ReLU gives linear results, GELU produces non-linear results



Gaussian Error Linear Unit (GELU)

Activation function to approximate a Gaussian cumulative distribution function (CDF): differentiable everywhere, enabling efficient gradient-based (= error) optimization

Standard Gaussian cumulative distribution function...

GELUs more easily approximate complicated functions than can ReLUs or ELUs. GELU has a probabilistic interpretation (Hendrycks & Gimpel, 2016)

input x ↓

$$\text{GELU}(x) = xP(X \leq x) = \frac{x\Phi(x)}{x\Phi(x)} = x \cdot \frac{1}{2} \left[1 + \text{erf}(x/\sqrt{2}) \right]$$

Standard Gaussian cumulative distribution function... $\Phi(x)$ ↑

erf = error function

↓ $x\Phi(x)$

...scales x by how much greater it is than other inputs

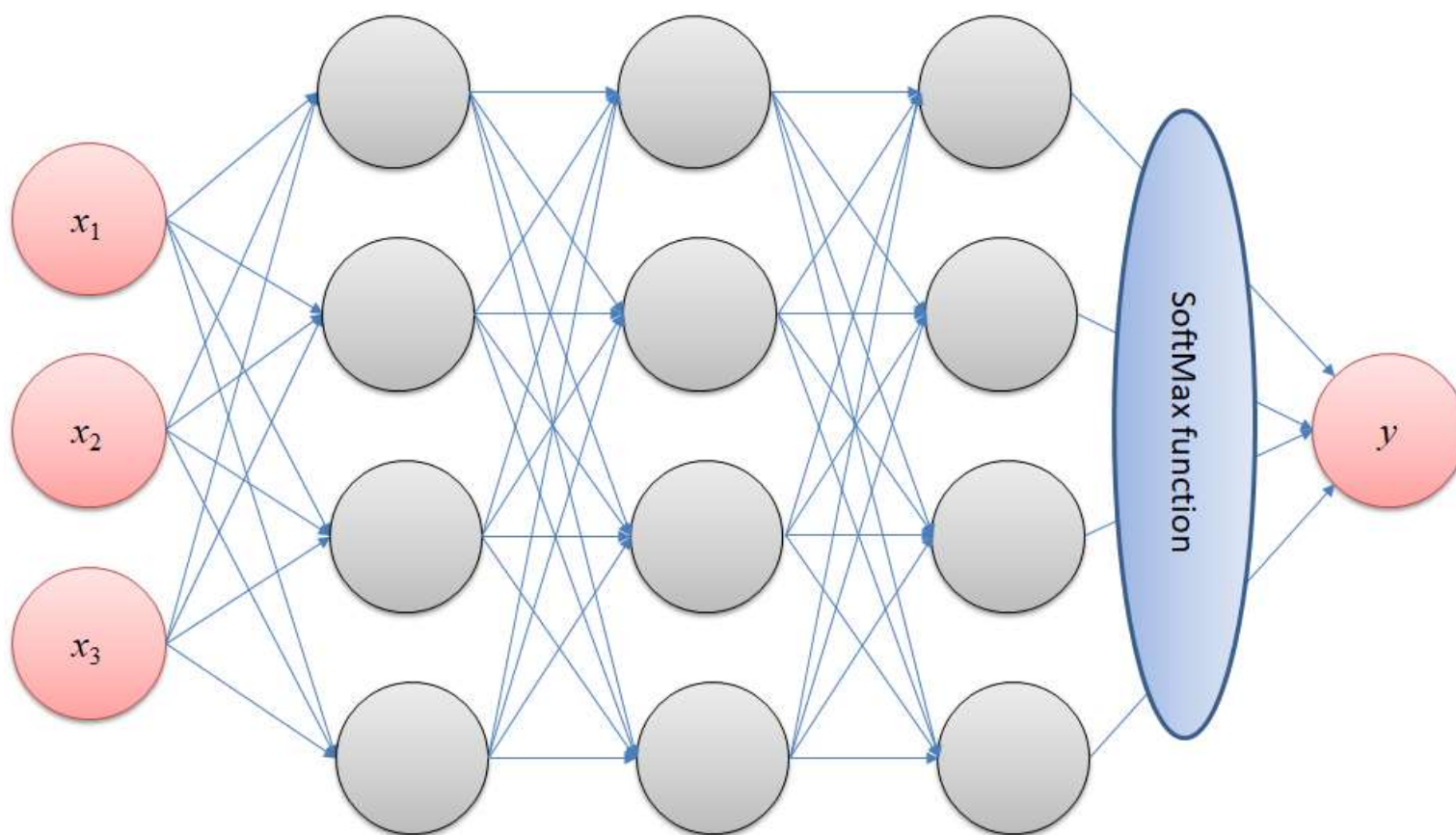
- ReLU ($x1_{x>0}$) linearly forwards inputs by their signs

- GELU's nonlinearity gives weights to inputs by their value

- Divide the input x by the square root of 2
- Apply the error function (erf) to the normalized input
 - Add 1 to the result of the error function
- Multiply the entire expression by $\frac{1}{2}$ to scale the output

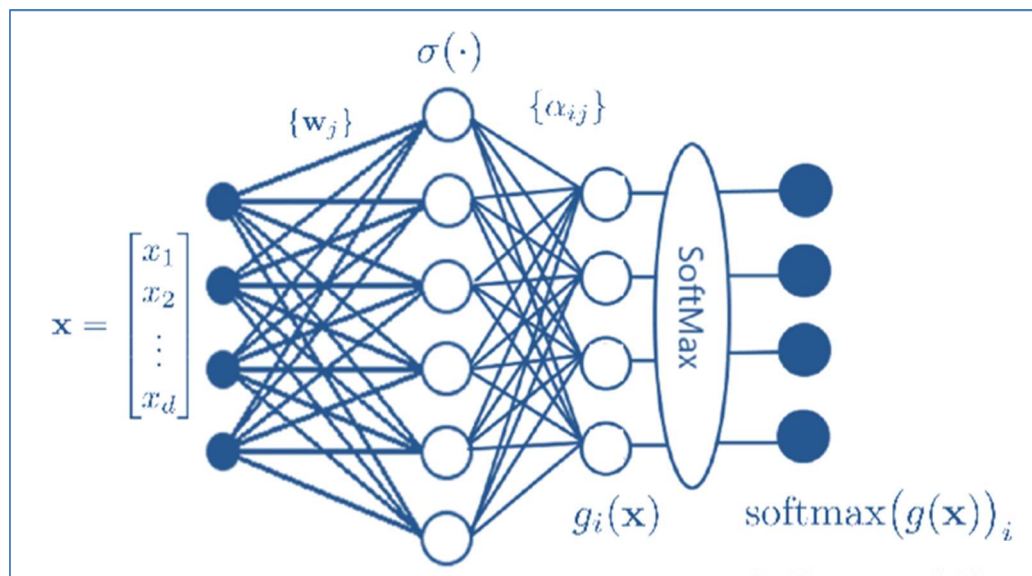
Softmax activation function for multi-class classification problems

The Softmax algorithm is the last station before producing an output and is to simplify the complexity of all those nodes firing $\text{weights} \times \text{features}$ at it, providing a probability distribution which features are most telling to identify the input is a cat and not a chair



Softmax activation function for multi-class classification problems

Softmax in the last layer is to simplify the results, mapping a vector of logits or real numbers (the input) to a vector of probabilities $\vec{z} = [z_1, z_2, \dots, z_n]$. Each probability ranges between 0 and 1 but such that the total probability sums up to 1. That way, the Softmax function offers a probability distribution



- Exponents are used to ensure that all probabilities are positive

- The denominator with the sum of the exponentiated input normalizes the probabilities to add up to 1

$$\text{Softmax } \sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

Annotations for the equation:

- e = exponent (points to the base e in the numerator)
- z_i (points to the exponent z_i in the numerator)
- Is input in class j ? (points to the index j in the denominator's summation)
- Sum of exponential parameters for all possible outcomes or classes (points to the denominator sum)
- $\exp(z_1) + \exp(z_2) + \dots + \exp(z_n)$ (points to the denominator sum)
- $\exp(z_i) /$ (points to the numerator)

Softmax is a generalization of the logistic function. Softmax gives you probabilities for each class: Decisions for a class are based on the highest probability. In contrast, Sigmoid is binary – two classes only: in or out

GPT is also Pre-trained by humans

GPT employed reinforcement learning from human feedback (RLHF)

- Trainers played the AI as well as the user, conversations were fed to the network
- Responses by the network were ranked by the trainers, which served as reward models, influencing the probability distributions towards more humanlike choices
- Today, users can upvote or downvote GPT's replies and give feedback to fine-tune results



GPT was trained on Microsoft's Azure supercomputers with NVIDIA GPUs for gaming

However, NLP by Azure Machine Learning: still no clue about the context



To: HOORN, Johan F [SD];

Start reply with: Congratulations! Good to hear! Great news!

Hi Johan,

I became a close contact, had lunch last Friday with a now confirmed case, my PhD student. I have been tested negative and doing well now, but I can't meet with people in a week time.

Best regards



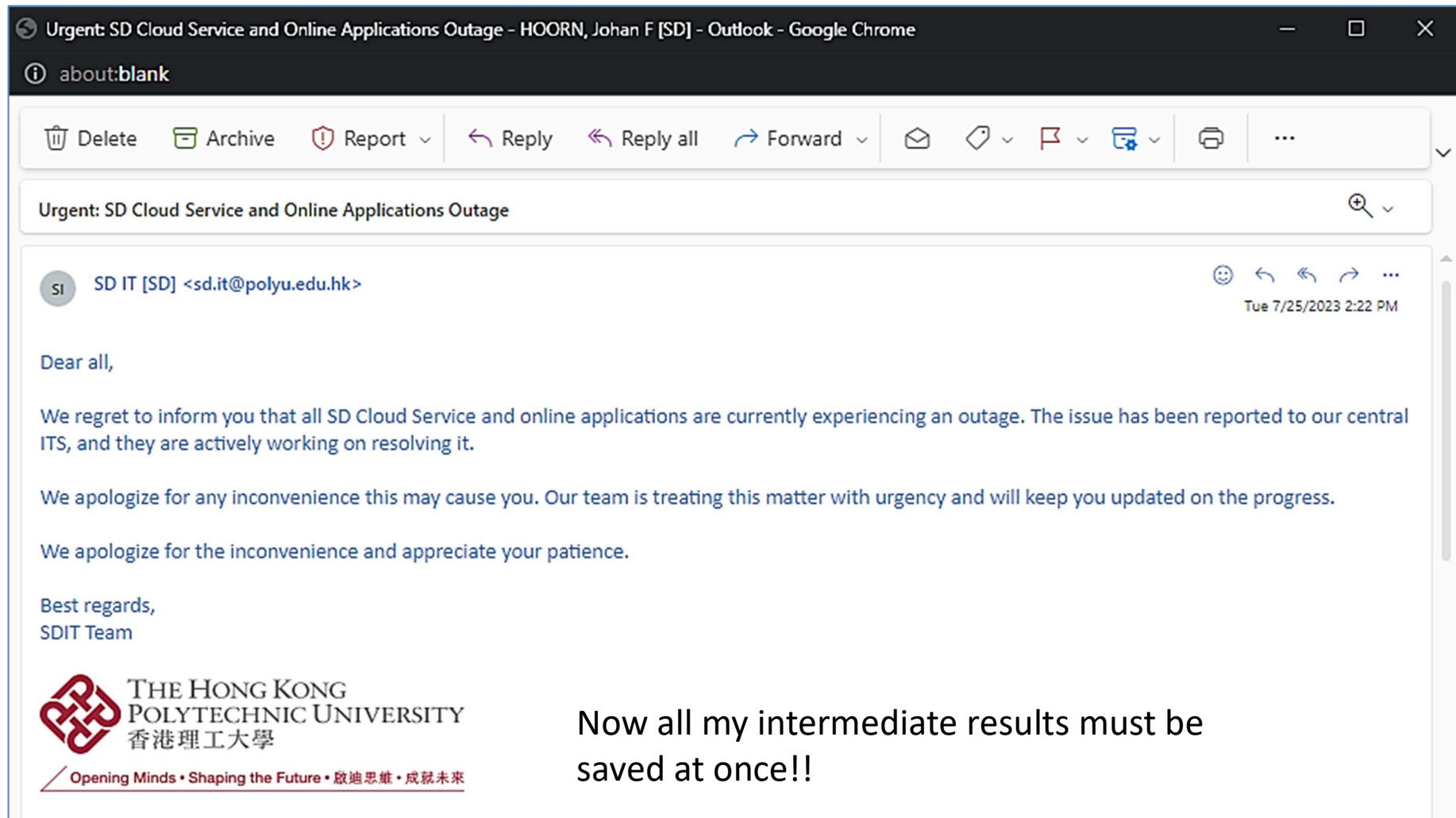
Wed 9/7/2022 9:48 AM

'I can't meet people for a week'

What shall I click?

Congratulations! Good to hear! Great news!

I was just training my model...



Hardware challenges for training LLMs

LLMs cost lots of time and consume lots of energy (not sustainable), because, according to NVIDIA (J. Lai):

- Massive data
- GPU utilization (of many GPUs)
- Inference
- Large systems (computing, communication, storage)
- Algorithms may be slow

How do hardware requirements relate to Human-Computer Interaction?

Infrastructure

Storage slow
Network slow
CPU slow
Limited bandwidth
Limited GPU memory

Engineering

Single threaded data load
Suboptimal decoding
Suboptimal augmentation
Synchronous execution of loading and compute
Lack of data queuing
Insufficient parallelism

Algorithms

Too small batch size
Computationally suboptimal network architecture
No TensorCore use to speed up processing without losing accuracy

Moral concerns: ChatGPT is immoral because humans are

ChatGPT's inconsistent moral advice influences users' judgment

Sebastian Krügel^{1✉}, Andreas Ostermaier² & Matthias Uhl¹

ChatGPT is not only fun to chat with, but it also searches information, answers questions, and gives advice. With consistent moral advice, it can improve the moral judgment and decisions of users. Unfortunately, ChatGPT's advice is not consistent. Nonetheless, it does influence users' moral judgment, we find in an experiment, even if they know they are advised by a chatting bot, and they underestimate how much they are influenced. Thus, ChatGPT corrupts rather than improves its users' moral judgment. While these findings call for better design of ChatGPT and similar bots, we also propose training to improve users' digital literacy as a remedy. Transparency, however, is not sufficient to enable the responsible use of AI.

What data do you train a network on?

Moral concerns: social bias

Whose Opinions Do Language Models Reflect?

Shibani Santurkar
Stanford

shibani@stanford.edu

Esin Durmus
Stanford

esindurmus@cs.stanford.edu

Faisal Ladhak
Columbia University

faisal@cs.columbia.edu

Cinoo Lee
Stanford

cinoollee@stanford.edu

Percy Liang
Stanford

pliang@cs.stanford.edu

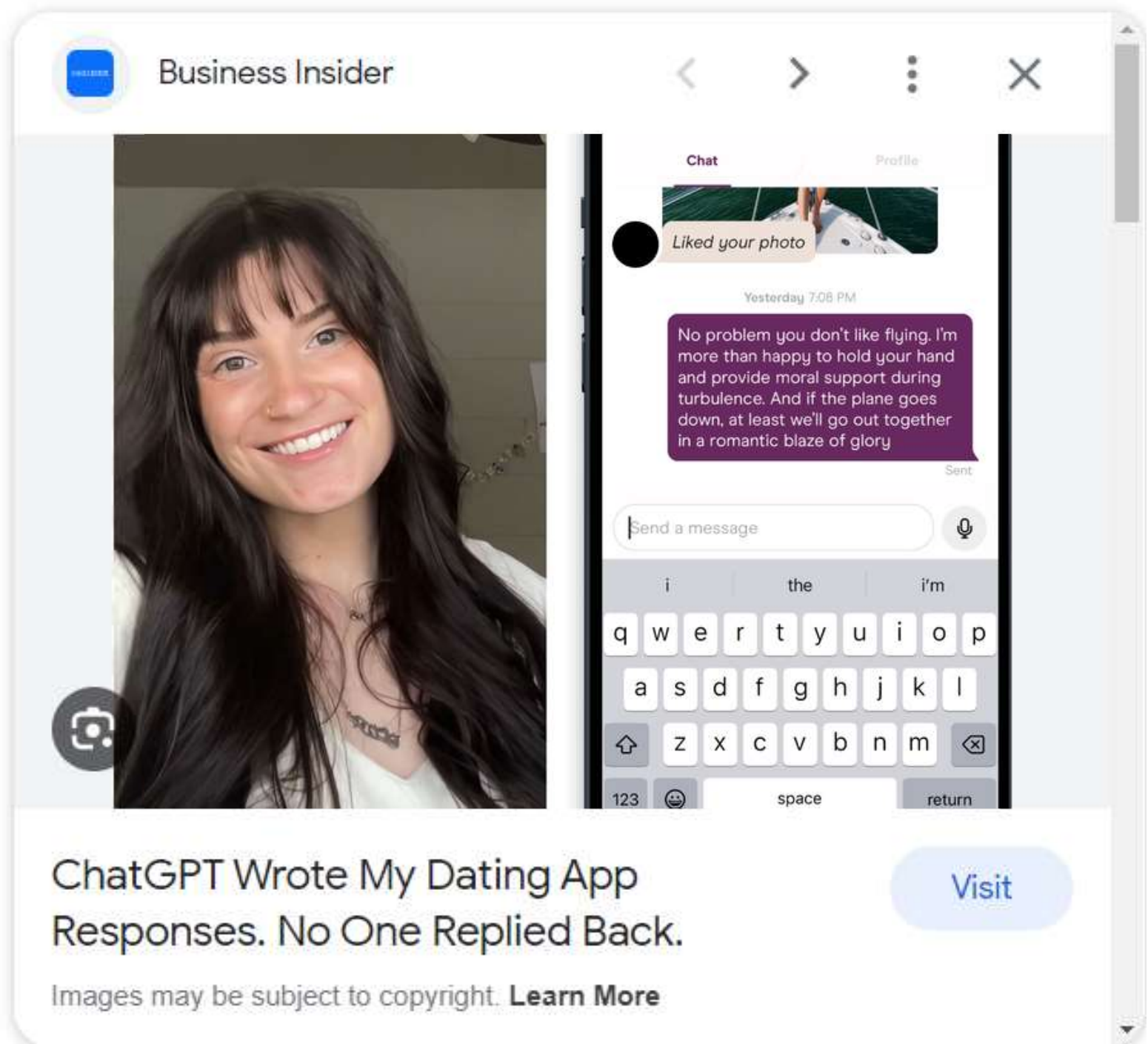
Tatsunori Hashimoto
Stanford

thashim@stanford.edu

Who are the people giving feedback?

Inappropriate
responses

Probability
estimates
allow
errors to
occur



5 minutes break

GPT Models Can Do More Than Just Talk: How to Make Them Brew You Coffee ☕ — Node.js



Raja Osama · Follow

Published in Artificial Intelligence in Plain English · 4 min read · 6 days ago



```
Make me a coffee => (User Provided Prompt = "Can you make me a coffee?") => Make  
Bring me a coffee => (User Provided Prompt = "Can you bring me a coffee?") => Ma  
Buy me a coffee => (User Provided Prompt = "Can you buy me a coffee?") => MakeAC
```

```
const intents = {  
  categories: {  
    "Make a coffee": MakeACoffee,  
    "Make me a coffee": MakeACoffee,  
    "Bring me a coffee": MakeACoffee,  
    "Buy me a coffee": MakeACoffee,  
  },  
};
```

```
const MakeACoffee = {  
  action: "LOCAL_ACTION",  
  response: () => "Your Espresso coffee is here!",  
};
```

```
const MakeACoffee = {  
  action: "API_CALL",  
  url: "http://localhost:3000/coffees",  
  ask: [  
    (coffees) => {  
      return `What kind of coffee would you like ${coffees.map(  
        (coffee) => coffee  
      )}?`;  
    },  
  ],  
  categories: () => ({  
    Mocha: MochaCoffee,  
    Karak: KarakCoffee,  
    Espresso: EspressoCoffee,  
  }  
),  
};
```

```
const MochaCoffee = {  
  action: "LOCAL_ACTION",  
  response: (response) => "Your mocha coffee is here!",  
};
```

ChaiGPT: Genuinely Pure Tea



facebook



I am Programmer,I have no life.

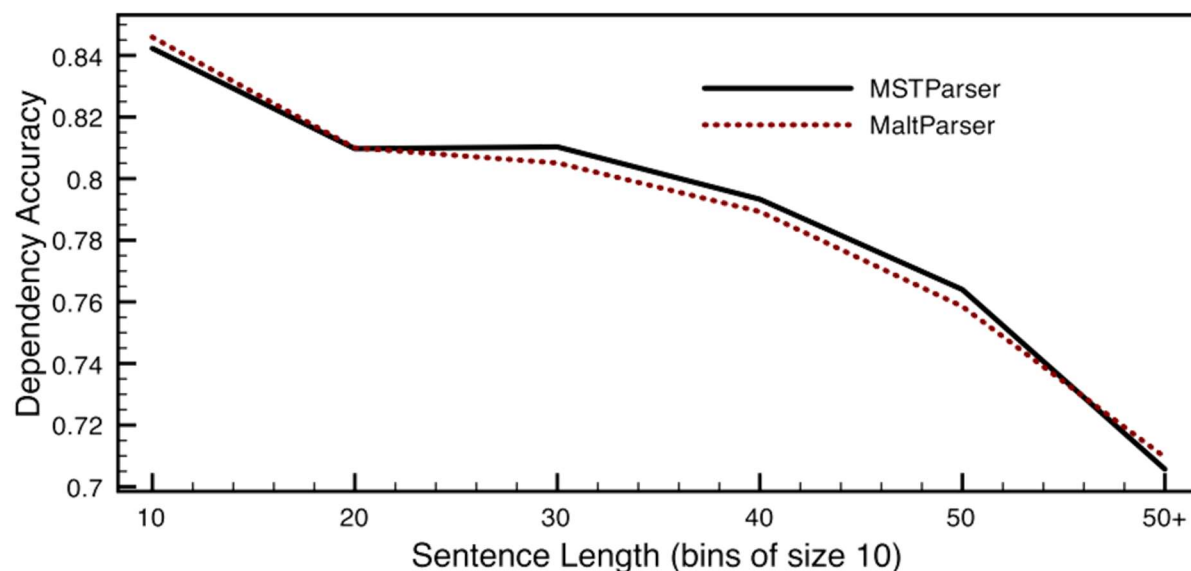
Tijdlijnfoto's · dinsdag om 11:44 ·

Machines and humans talking together

□ Natural Language Processing (NLP)

Difficulties with NLP

- Error Cascading
 - Processing natural language is a multi-step process
 - Errors at one stage lead to more errors further down



- Accuracy drops with longer sentences

Machines and humans talking together

❑ Natural Language Processing (NLP)

Difficulties with NLP

- Ambiguity

Ambiguous Words	Different Meanings	Number of Different Possible Meanings
catch	a game, haul, stop	thirty-nine
return	repeat, income	twenty-nine
well	healthy, good, source	twenty-two
bar	to ban, salad bar, legal profession	nineteen

Machines and humans talking together

❑ Not just in Natural Language Processing (NLP)

- Ambiguity in coding

```
// btnConnectArduino
//
this.btnConnectArduino.Font = new System.Drawing.Font("Microsoft Sans
Serif",
7.8F);
this.btnConnectArduino.ForeColor = System.Drawing.Color.FromArgb(((int)
(((byte)(0)))), ((int)(((byte)(13)))), ((int)(((byte)(51)))));
this.btnConnectArduino.Location = new System.Drawing.Point(6, 89);
this.btnConnectArduino.Name = "btnConnectArduino";
this.btnConnectArduino.Size = new System.Drawing.Size(90, 25);
this.btnConnectArduino.TabIndex = 2;
this.btnConnectArduino.Text = "Connect";
this.btnConnectArduino.UseVisualStyleBackColor = true;
this.btnConnectArduino.Click += new
System.EventHandler(this.btnConnectArduino_Click);
//
private void btnConnectArduino(object sender, EventArgs e)
{
//Processing serial port to connect reader.
string strException = string.Empty;
string strComPort = cmbComPort.Text;
int nBaudrate = Convert.ToInt32(cmbBaudrate.Text);
int nRet = reader.OpenCom(strComPort, nBaudrate, out strException);
if (nRet != 0)
{string strLog = "Connection failed, failure cause: " + strException;
WriteLog(lrtxtLog, strLog, 1);
return;
}else
{
```

button

method

Machines and humans talking together

- ❑ Not just in Natural Language Processing (NLP)
- Ambiguity in visual design



Teiyu Goto, designer of PlayStation controller

Machines and humans talking together

- ❑ Not just in Natural Language Processing (NLP)
- Ambiguity in visual design



Reversed in
Western
games!

No, cancel

Yes, confirm

Teiyu Goto, designer of PlayStation controller

Machines and humans talking together

- ❑ Not just in Natural Language Processing (NLP)
- Ambiguity in visual design

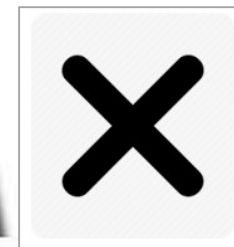


Reversed in
Western
games!

Could also
mean STOP



Could be a cancel
button yet



Teiyu Goto, designer of PlayStation controller

Machines and humans talking together

❑ Not just in Natural Language Processing (NLP)

- Ambiguity in visual design



Point of view
or
Moving up
or
Moving forward
or
Warning

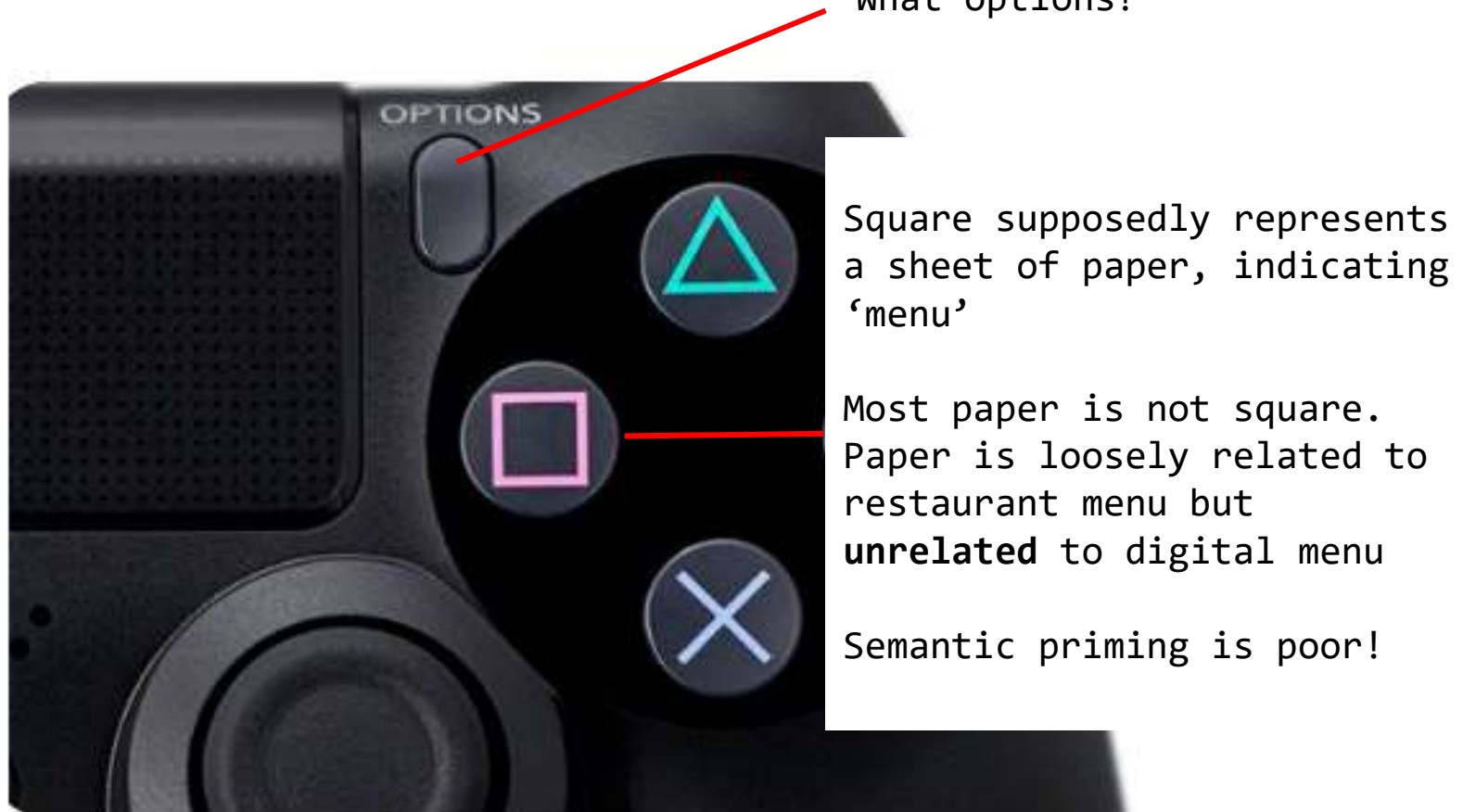


Teiyu Goto, designer of PlayStation controller

Machines and humans talking together

❑ Not just in Natural Language Processing (NLP)

- Ambiguity in visual design



Teiyu Goto, designer of PlayStation controller

Machines and humans talking together

❑ Natural Language Processing (NLP)

The PS controller shows that language (GPT) is not enough to enable Human-Computer Conversation

- It misses pragmatics, situational awareness
- Multimodal input: What does *this* button mean?



said with
emphasis

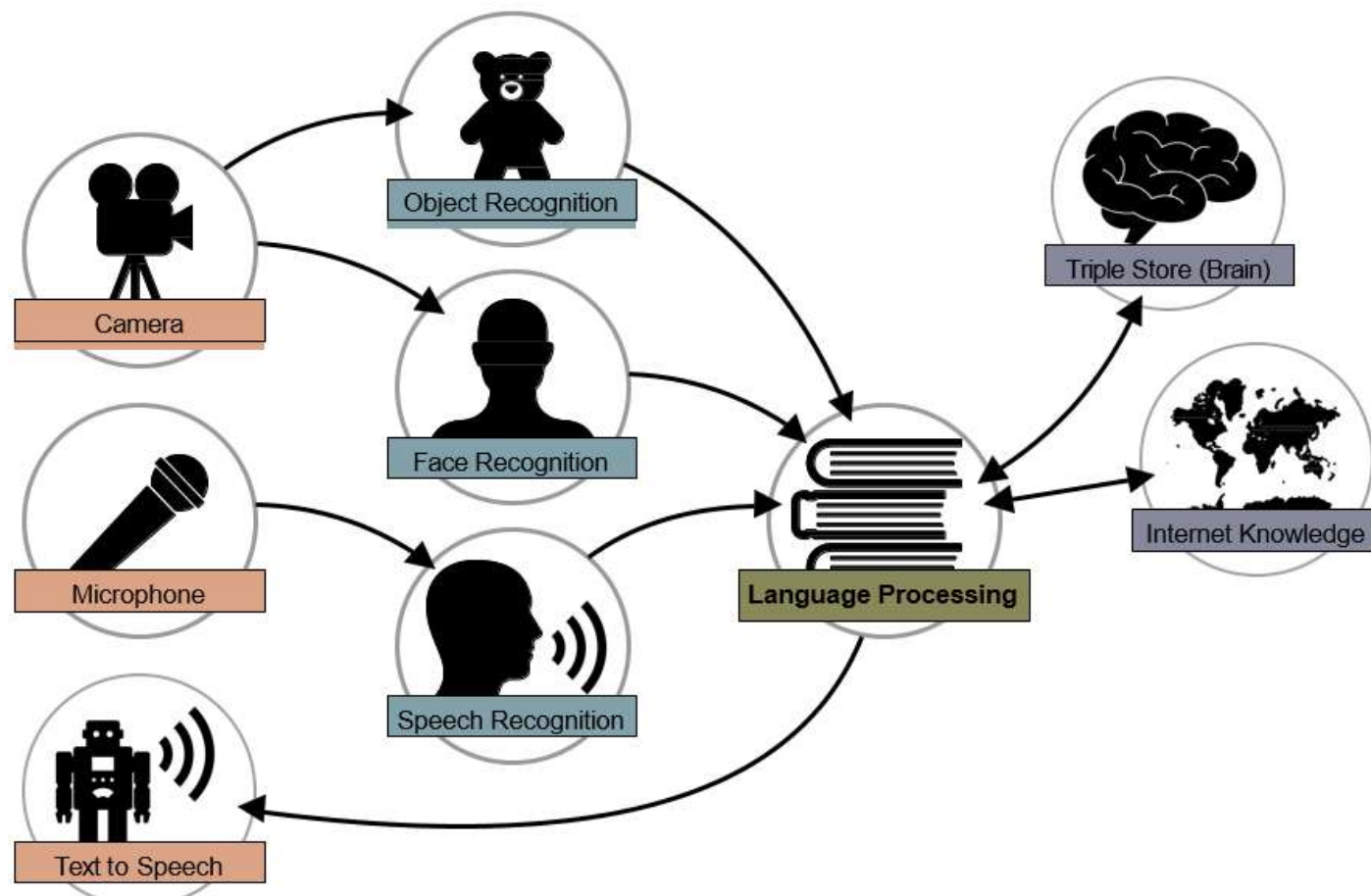


pointing
with finger

Machines and humans talking together

❑ Natural Language Processing (NLP)

- Multimodal input: Enabling Human-Computer Conversation



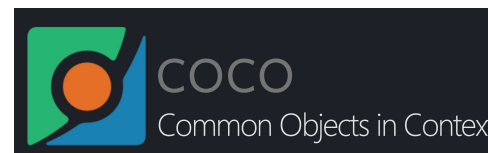
Machines and humans talking together

❑ Natural Language Processing (NLP)

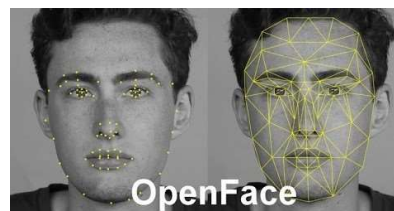
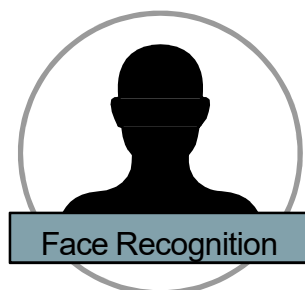
Language is not enough! Pragmatics, situational awareness

- Multimodal input: I point and say “This button, that menu”

Object
recognition



Face detection



Detecting and
recognizing
speech

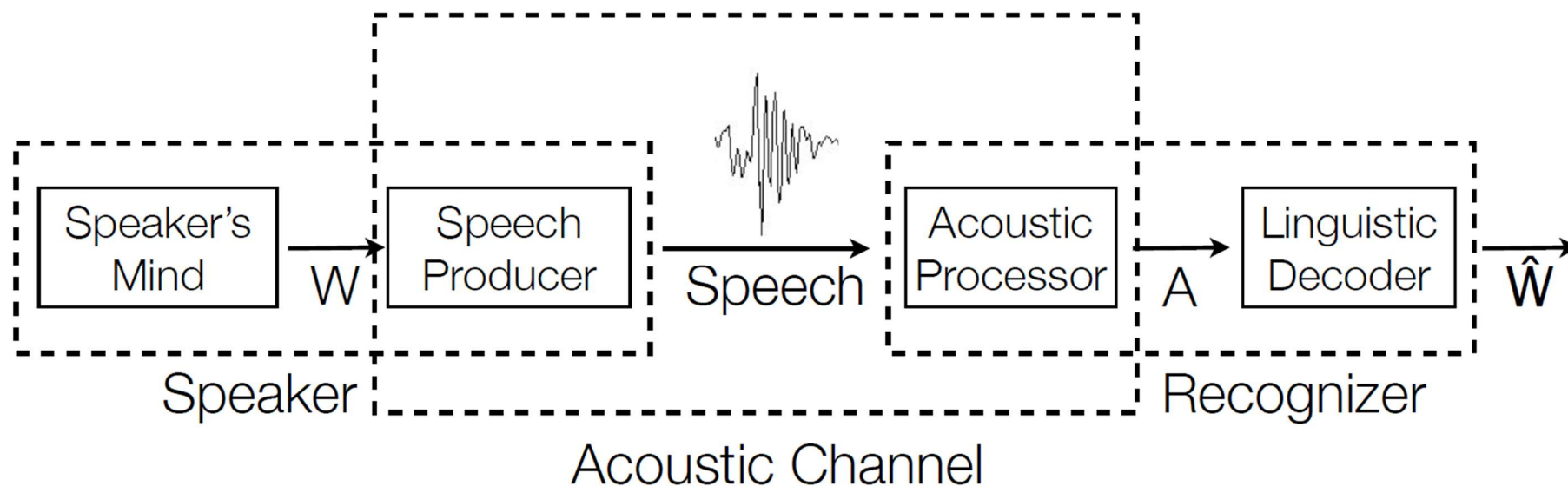


Machines and humans talking together

□ Natural Language Processing (NLP)

Speech Recognition (two forms)

- Converts spoken words to machine-readable input (text)
- Identifies the speaker according to acoustic input (voice verification)



Machines and humans talking together

□ Natural Language Processing (NLP)

Speech Recognition Methods

- Before, techniques mostly came from Electrical Engineering and Mathematics:
 - Digital Signal Processing
 - Probabilities and Statistics, Modeling
- Measured in Word Error Rate
(number of wrong words/length of sentence)
- Problems:
 - Speaker dependence
 - Acoustic sampling
 - Environmental noise

Machines and humans talking together

□ Natural Language Processing (NLP)

As said, language is not enough! Situated context

- But image recognition only gives the type of object and not the instance
- Identifying object instances is difficult but needed:
 - count multiple objects (chairs), identify object across contexts, assign properties to different instances of objects (my chair is broken), object instances may move, disappear, appear, etc.

Extreme solutions:

- All objects of the same type (e.g., chairs) are the same instance (chair)
- Every perception of an object results in a new instance (as many chairs as there are perceptions)

Machines and humans talking together

❑ Natural Language Processing (NLP)

Match places and objects within contexts

Location defines object instances and objects define the location:

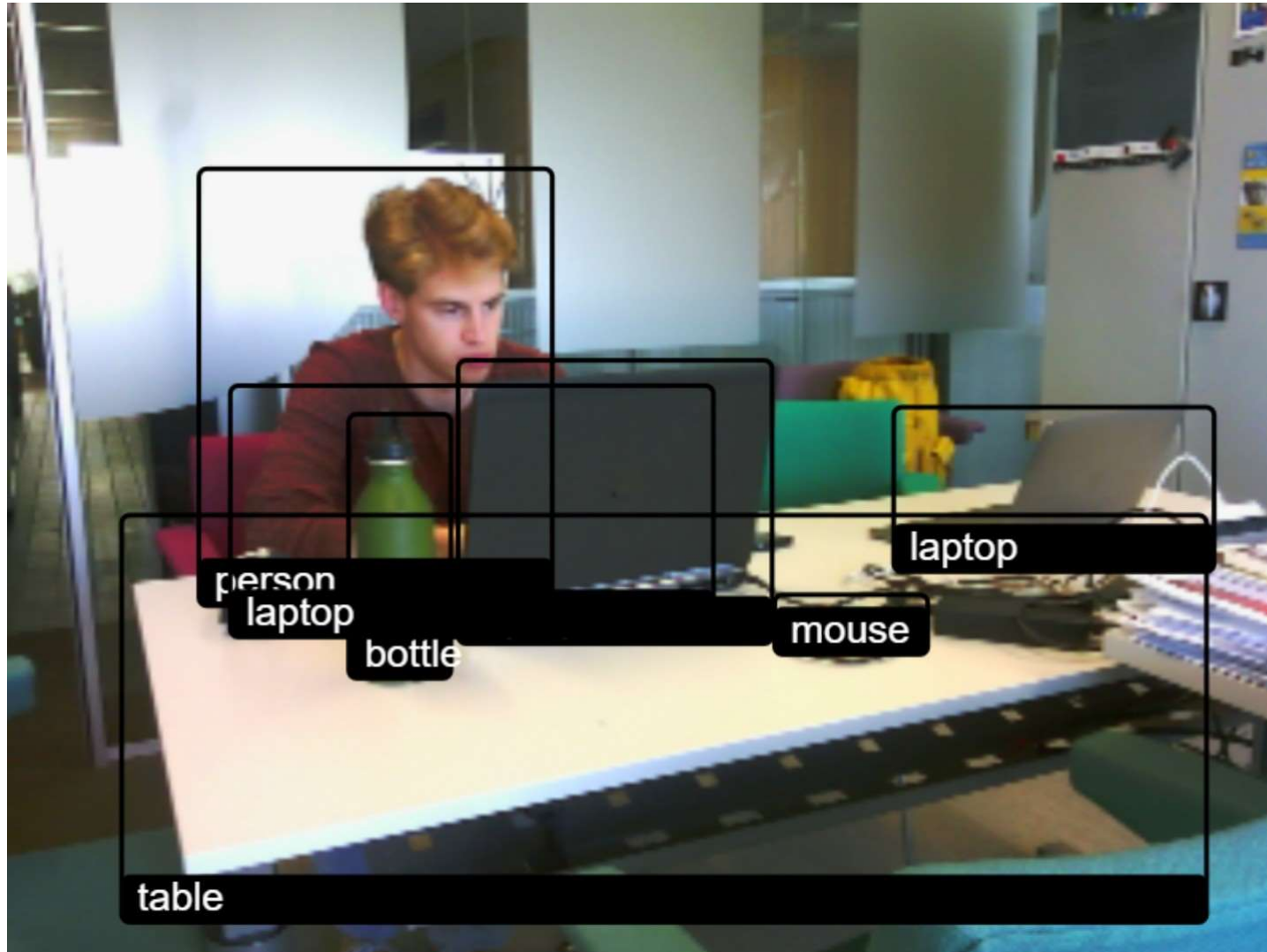
- Objects in known locations mapped to known object instances
- Objects in different locations mapped to different instances

Differentiate objects when they do not match in properties (size, color, position, ownership):

- Permanence: How much can change to still identify an object?
- Changeability of properties of things: movable, transferable, size, color

Machines and humans talking together

- ❑ Natural Language Processing (NLP): Context awareness



Machines and humans talking together

- ❑ Context awareness in NLP: spatial map to locate objects



Machines and humans talking together

- ❑ NLP: categorization through episodic experiences

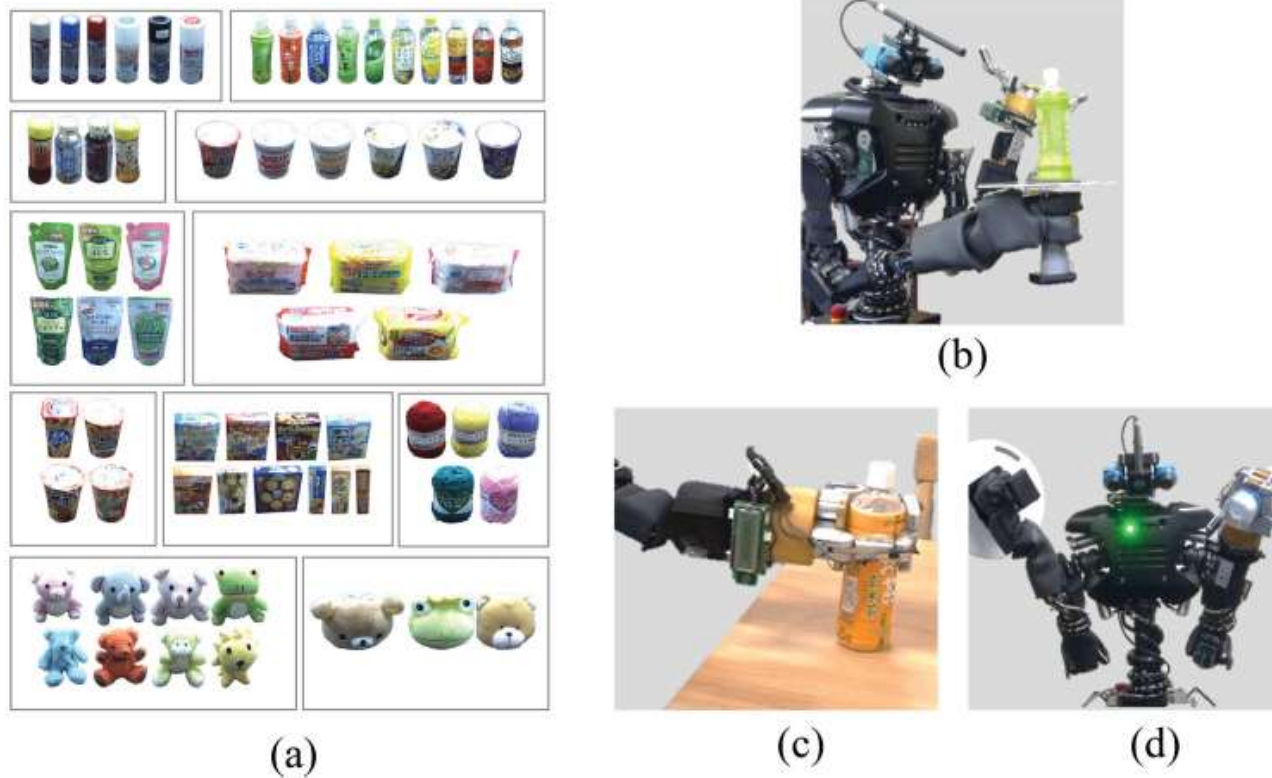


Figure 2. Object concept formation in a robot: (a) objects used in the experiment, and obtaining visual, haptic, and auditory information by (b) observing, (c) grasping, and (d) shaking objects.

Tangiuchi et al. 2019

Machines and humans talking together

□ NLP: hierarchical clustering

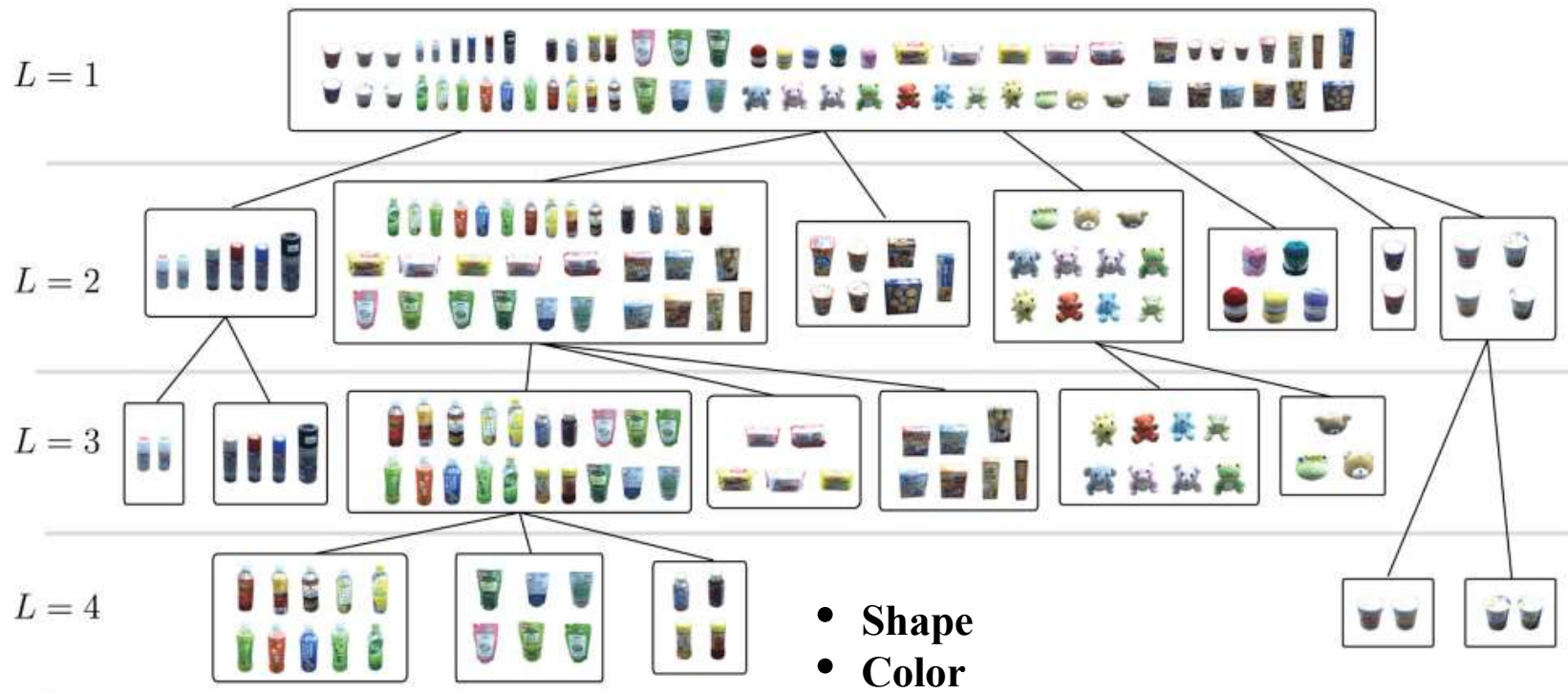
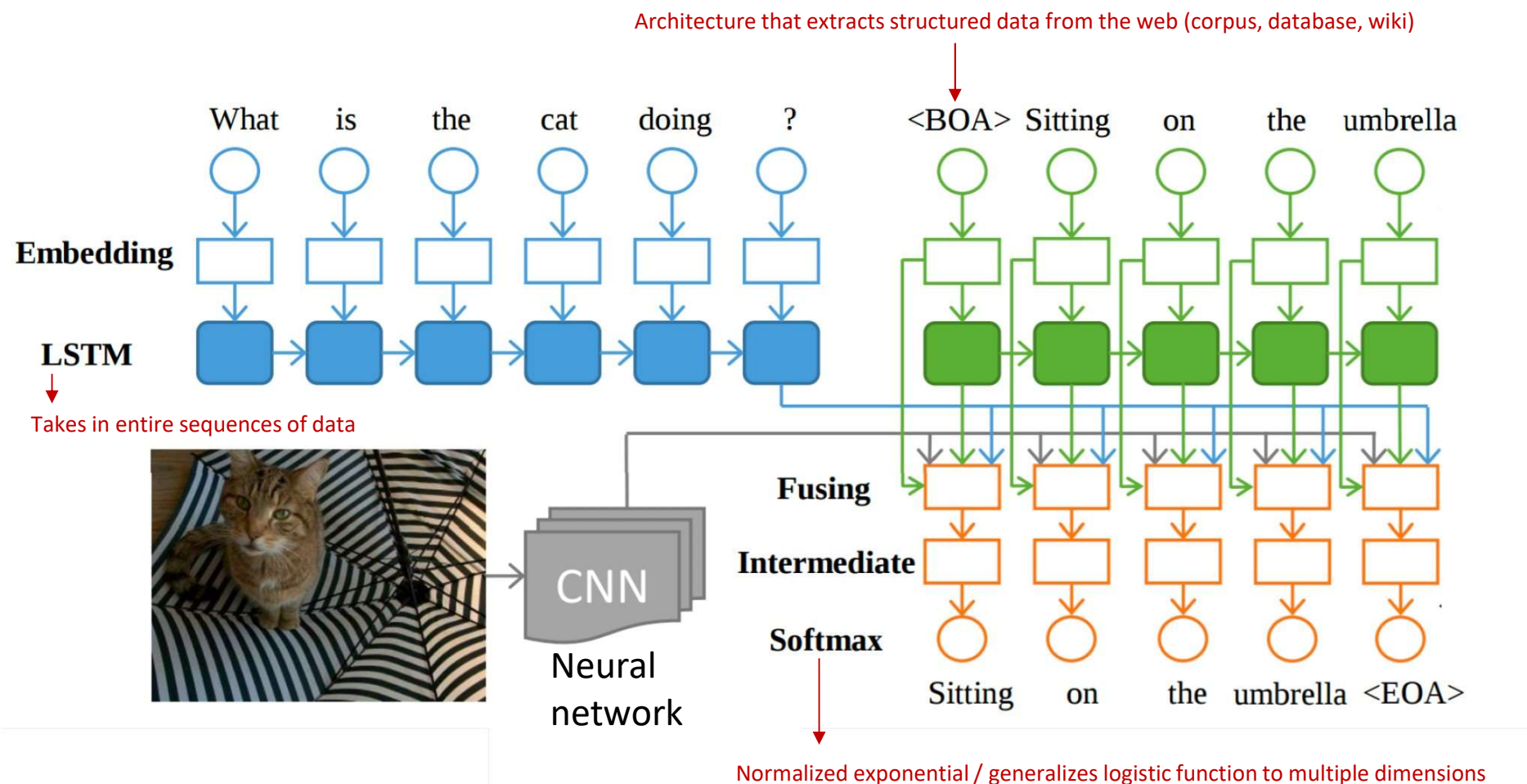


Figure 4. Hierarchical concepts formed by robot.

- Shape
- Color
- Function, affordances
- Weight, texture, smell, sound, solidness
- Contexts, dynamics, action-states
- Reference through language

Machines and humans talking together

□ NLP: multimodal learning



Sample application: Grounding action verbs

NLP has problems with interpreting a situation due to ambiguity of meaning. Here, action verbs (e.g., make me a tea) are connected to the situation by reinforcement learning, letting the system ask questions that the human answers and relating that information to multimodal input. System estimates the most likely meaning of the verb



**Humans
interpreting
machines**

Humans interpreting machines

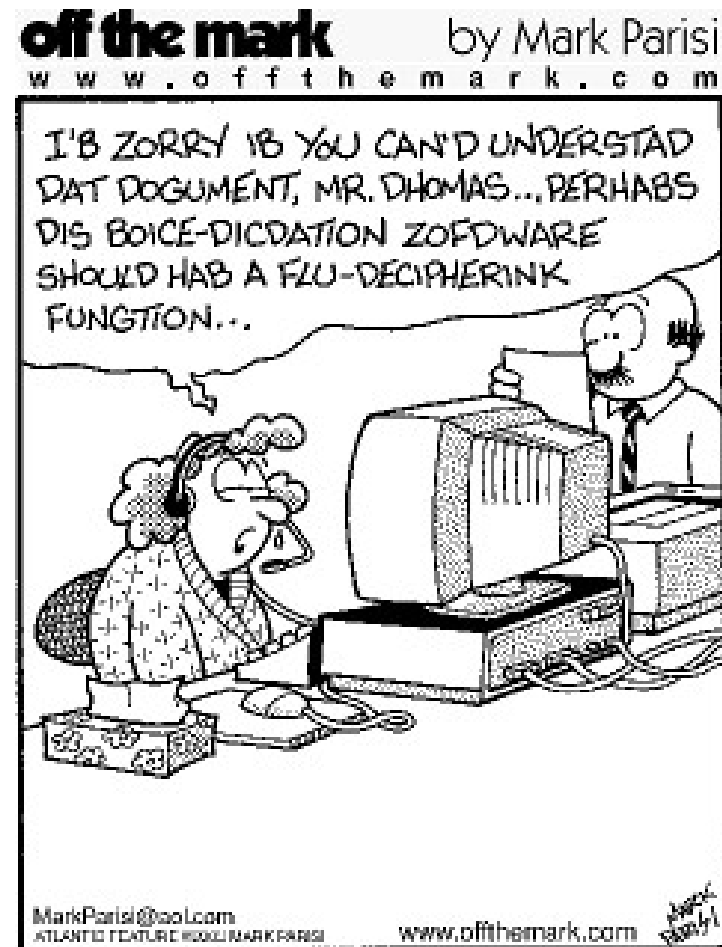
Computer needs literal instruction

```
jfhoom@flits:/home/jfhoom
Sun Microsystems Inc.  SunOS 5.8
$ password
bash: password: command not found
$ passwd
passwd: Changing password for jfhoom
Enter login(NIS+) password: █
```

Although <password> is unambiguous language, the Unix shell can only interpret the abbreviation <passwd>

Phonemes should be intact

Human gets the gist

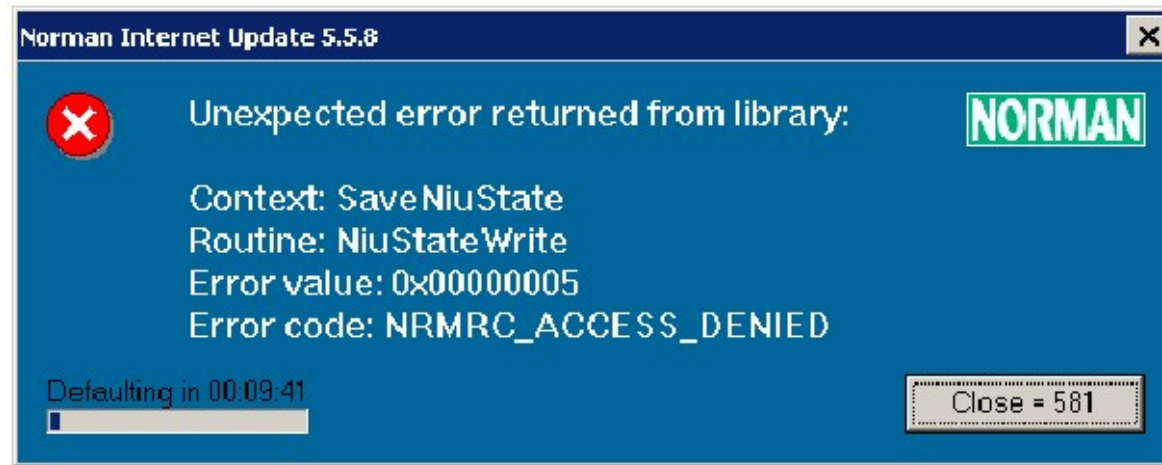


Retrieved March 18, 2003 from <http://www.viewz.com/cartoon/cartoon.shtml>

Phonemes can be degraded 81

Humans interpreting machines

Then why is this text not fit for human consumption?



This is literal computer language. For humans, there is no gist to get, except for "something is wrong but I don't know what," which makes this message unnecessary worrying

Thus, human context $\neq$ computer context (cf. SaveNiuState)

For computers, grammar is crucial

For humans, semantics dominates grammar

Proper Python syntax:

```
if (<boolean expression>){  
    S1;  
} else {  
    S2;  
}
```

If this is true,
do the following.
Otherwise, do something else.

Problematic order of
conditional/statements:

```
{  
    S1;  
if (<boolean expression>)  
    } else {  
    S2  
}
```



Incorrect interpunction is fatal

Although grammatically unusual, this is a
semantically unproblematic order of
conditional/statements:

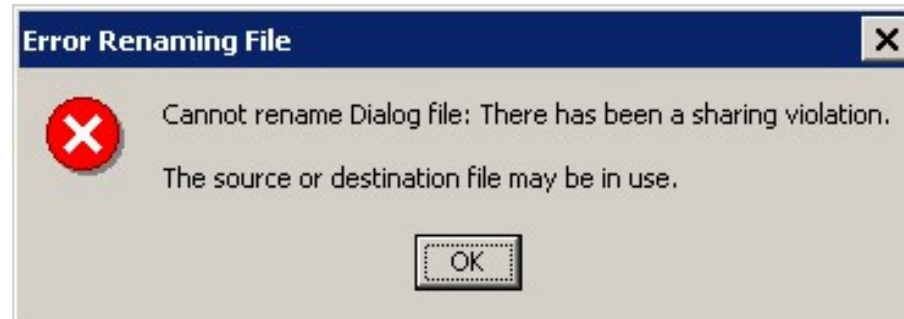
Do the following,
if this is true.
Otherwise do something else.



Incorrect interpunction is subordinate

Semantics can topple grammar, due to human interpretation in context

Someone renames an open file. The warning is grammatically correct:



However, users may not know exactly what a sharing violation is.

They think: "I am working on this file alone. What is the source file and what the destination file? I am working on my own file."

From a system perspective, indeed, the event is identical to a second user who tries to change a fixed pointer - the file name. Renaming is to the computer the same as moving one file (source) to another (the destination) because the fixed pointers suggest this. Because a computer cannot interpret contents (it is the same doc) or context (the same user works on it), it sees any file with a different name as a completely different file.

What are the pros and cons of focus on absolute differences (computer) and of a fuzzy similarity focus (human)?

← adaptability

← exactness

Humans interpreting machines

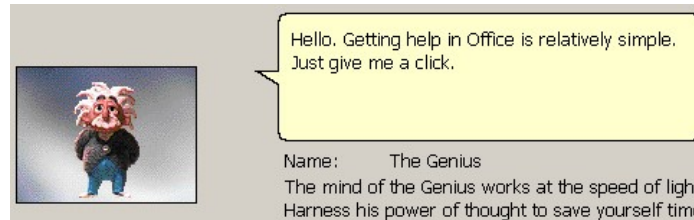


Thus, computer needs literal instruction,
human gets the gist

Computer needs correct grammar (cf. prescriptive grammar)
For humans, grammar is subordinate to
semantics (cf. descriptive grammar)

Computer is capable of context free interpretation
Human performs context-sensitive interpretation (social, semantic)

Speech acts



examples of context

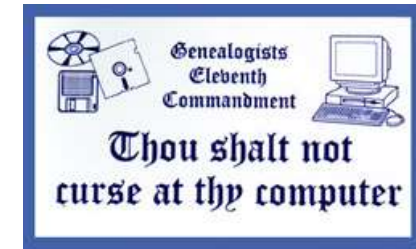
Representative Conveying a belief about truth

Directive Attempt to persuade the receiver

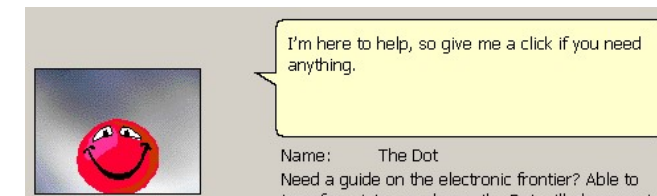
Commissive Promise to engage in future action

Expressive Stating a psychological state

Declarative Stating a new state of affairs



Retrieved August 24, 2003 from <http://www.petersenprints.com/store/media/magnet-computer.jpg>



Retrieved August 26, 2003 from <http://stairway.org/sorry.html>

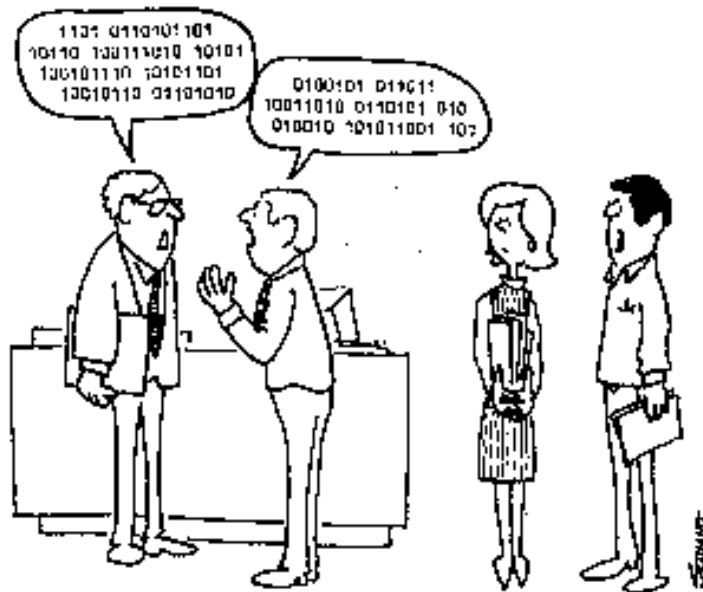
create module calendar

```
% A class to store dates
declare date ->> entity
declare day(date) -> string
declare month(date) -> string
declare year(date) -> integer
```

Retrieved August 24, 2003 from <http://www.computer.org/proceedings/meta/1999/papers/62/sembury-img2.gif>

Conversation rules

Quantity	Provide the right amount of information (no more no less)
Quality	Give an honest rendition of your beliefs (tell the truth)
Relation	Information should be relevant to the goals of a conversation
Manner	Be clear to your audience (no jargon, no beating about the bush)



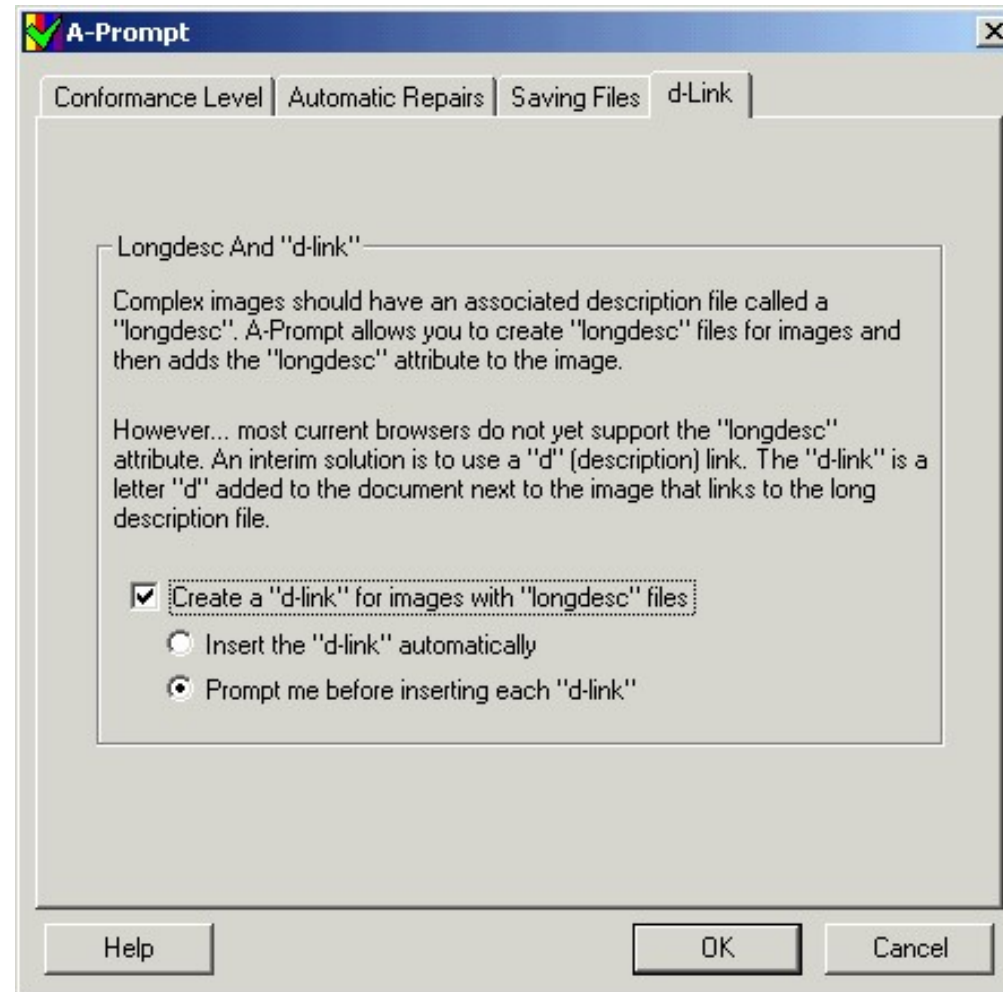
Drawn for BROADCASTING by Jack Schmidt

"I hate it when engineers talk shop."

Retrieved August 24, 2003 from <http://www.mwpersons.com/Jokes.html>

Quantity

Provide the right amount of information (no more no less)



Quality

Give an honest rendition of your beliefs (tell the truth)

[Symantec Security Response - Hoax Page](#)

... Please disregard the **hoax** emails - they contain bogus warnings usually intent only on frightening or misleading users. The best ...

Beschrijving: Uncovering hoaxes that usually arrive in the form of an email, or promotions that sound too good to be true.

Categorie: [Society > Folklore > ... > Urban Legends > Computer Virus Hoaxes](#)

www.symantec.com/avcenter/hoax.html - 17k - [In cache](#) - [Gelijkwaardige pagina's](#)

[HOAXBUSTERS Home Page](#)

... Welcome to the CIAC **Hoax** Pages. Last modified: 07/14/03 You are the 13108433rd visitor to the CIAC Internet **Hoax** and Chain Letter pages since February 1995. ...

Beschrijving: Information about hoaxes, chain letters, urban myths and other bogus information being routed around...

Categorie: [Society > Folklore > Literature > Urban Legends](#)

hoaxbusters.ciac.org/ - 9k - [In cache](#) - [Gelijkwaardige pagina's](#)

[McAfee.com - Virus Hoaxes](#)

... For example, AOL4FREE began as a **hoax** virus warning. Then somebody distributed a destructive trojan attached to the original **hoax** virus warning! ...

vil.mcafee.com/hoax.asp - 46k - [In cache](#) - [Gelijkwaardige pagina's](#)

Relation

Information should be relevant to the goals of a conversation

 Dit is de afbeelding: developer.iplanet.com/viewsource/gee_nas/cookie.GIF



[Frame weghalen](#)
[Terug naar resultaten](#)

*Ware grootte. [Bekijk slechts de afbeelding.](#)
Deze afbeelding kan auteursrechtelijk zijn beschermd.*

Hieronder vind je de afbeelding in zijn oorspronkelijke context op de pagina: developer.iplanet.com/viewsource/ae_nas/ae_nas.html

developers.sun.com

 **The Source for Developers**
» Products & Technologies » Technical Topics



[Developer Home >](#)[Profile and Registration | Why Register?](#)

Sun Microsystems, Inc. has launched the Sun Developer Network

You arrived at this page because we could not find the page you requested from one of the consolidated sites. If you would like to learn more about why Sun is reworking its developer sites, continue reading, OR,
→ [Get Started Now on the Sun Developer Network](#)

Sun Microsystems, Inc. has launched the Sun Developer Network, and has kicked off a project to consolidate more than 18 different technology sites into one site for developers. We are planning a phased release of the migrated sites and want to let you know what's changing, what to expect, and when to expect these changes.

Why is Sun reworking its developer sites?

Because we heard you. We met with thousands of developers around the world in the last year, and they voiced strong opinions about our Web sites. They told us that it's hard to use our existing set of distributed sites, and hard to find the content they need across multiple sites and technologies. To address these issues we have undertaken this consolidation project.

We have set these goals for the new consolidated developer site:

Create a one-stop destination for developers

We're consolidating and streamlining more than 18 different Sun Web sites so you can more easily find content, tools and answers. We know that most technical solutions require more than one technology, so our goal is to provide all technical content, tools, and community features in an integrated navigational scheme. Developers will have just one URL to remember, with easy access to integrated navigation and search functionality.

We're calling this site "The Source for Developers" because that's what we aim to create.

Make it easier to find, use and share Sun technology

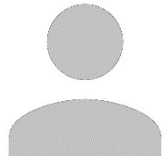
We are making the new site simple and consistent, enabling developers to quickly find the information they need. We have developed an intuitive interface and improved site tools, simplifying access to content for all Sun technologies. On the backend we are migrating all of these resources into a centralized content management system. Once this is complete, developers will be able to search and browse through documentation, tips and tutorials without having to navigate multiple Web sites.

Strengthen developer communities

Sun Microsystems' commitment to software developers is fundamental to our reputation for forward-thinking research and development. We know that establishing communication between developers can increase their effectiveness and the pace of innovation. So we are improving the integration between forums and content, to support communication about specific articles and code samples. We are also integrating project-based collaboration and WIKI functionality.

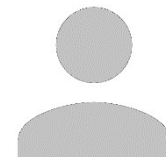
Relation

Information should be relevant to the goals of a conversation



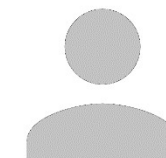
My students are depressed.

I'm really sorry to hear that. It's important to approach this situation with sensitivity. Try to encourage them to speak about their feelings and let them know it's completely okay to ask for help. If possible, refer them to a school counselor or a mental health professional. It might also be helpful to involve their parents or guardians if it feels necessary. If they are in immediate danger, such as expressing suicidal or self-harm behaviors, you should contact the necessary authorities or hotlines for help.



I mean, they don't look forward to the exam.

Ah, I see. That's also a common thing among students. It could be due to anxiety or fear of failure. Here are a few tips you can give them: 1. Plan Out Study Schedules: Regular, focused studying with ample breaks will help retain information better than last-minute cramming. 2. Exercise: Even light exercise can help reduce stress and improve mood and focus. 3. Stay Positive: Encourage them to visualize success, maintain a positive attitude and remind them that grades don't define their worth. 4. Study Groups: Group studies can make revisions more interactive and enjoyable. 5. Seek Help: If they're struggling with a particular topic, encourage them to seek help from teachers, tutors, or classmates. 6. Use Study Tools: Visual aids (like mind maps, flowcharts), flashcards, online quizzes can make studying more engaging. 7. Practice with Past Papers: This will help them understand the format and type of questions to expect, thus reducing anxiety. Lastly, remind them that it's okay to feel nervous before exams. What's important is to not let it overcome their preparations.



I do not seek advice. I just wanted to tell.

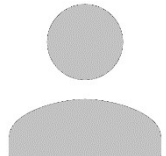
I'm really sorry that you're feeling this way, but I'm unable to provide the help that you need. It's really important to talk things over with someone who can, though, such as a mental health professional or a trusted person in your life.



0 / 10000

Relation

Information should be relevant to the goals of a conversation

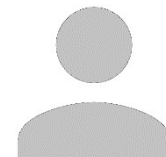


My students are depressed.

I'm really sorry to hear that. It's important to approach this situation with sensitivity. Try to

/* GPT4 has no idea about pragmatics: 'depressed' does not really mean clinically depressed */

danger, such as expressing suicidal or self-harm behaviors, you should contact the necessary authorities or hotlines for help.



I mean, they don't look forward to the exam.

Ah, I see. That's also a common thing among students. It could be due to anxiety or fear of failure. Here are a few tips you can give them: 1. Plan Out Study Schedules: Regular, focused

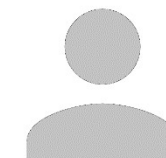
/* I wasn't seeking any advice, this is a weird assumption. I know better how to guide my students than GPT, really */

don't define their worth. 4. Study Groups: Group studies can make revisions more interactive and enjoyable. 5. Seek Help: If they're struggling with a particular topic, encourage them to seek help from teachers, tutors, or classmates. 6. Use Study Tools: Visual aids (like mind maps, flowcharts), flashcards, online quizzes can make studying more engaging. 7. Practice with Past Papers: This

/* GPT4 does not understand conversational goals/needs. This is just to share my experiences. But GPT does not check conversational goals, it just barges in and marches on */

I do not seek advice. I just wanted to tell.

/* Main issue: GPT does not check whether it understands things correctly. Runs from general and standard assumptions. No Q&A before it comes up with its 'definite' answers */

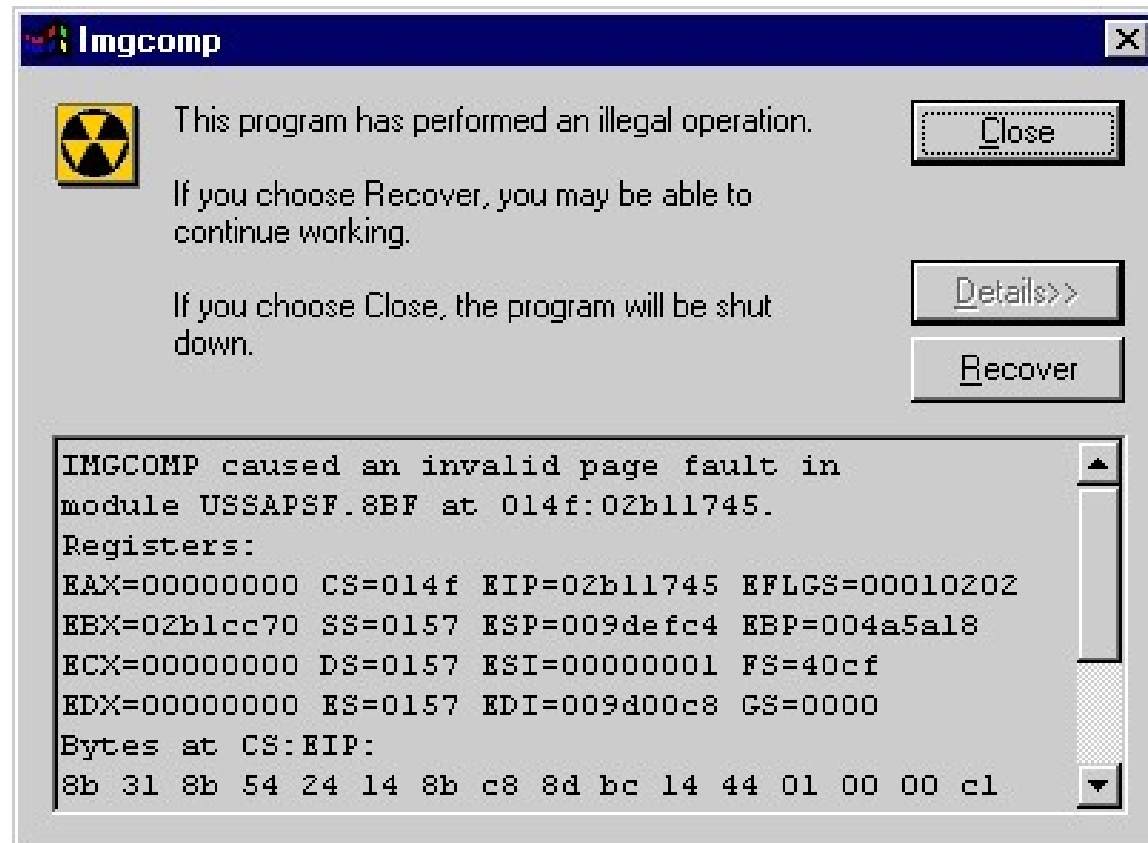


0 / 10000



Manner

Be clear to your audience (no jargon, no beating about the bush)



Retrieved August 27, 2003 from <http://www.pioneeris.net/ashton/IC/images/error.gif>

Scripts

Humans interpreting machines

Humans behave and thus, use their language, according to scripts; relatively fixed sequences of actions and behaviors associated with performing a task or achieving a goal.

For example, going to a library may have the following script:

Go in, consult catalogue or librarian, look up book, take book from shelf, have book checked out, put book in bag, go out

At each step, subscripts may come into action. For instance, in case the door is jammed or in conversing with the librarian

In system design, psychological scripts are the basis for scenario development

Imagine a human conversational script for getting assistance and compare it with Clippit's conversational 'skills'

Human conversational script: *Opening dialog*

Librarian: "Can I help you?"

Not: "What would you like to do" because perhaps I don't want to do anything, but just want to get informed



Human conversational script: *Pose question*

Visitor: "I am looking for Schank and Abelson, 1977"

Not: "Type your question here, and then click Search."

What misses: Simulation of conversational rules. This is a search engine dressed up like a humanoid

Microsoft Word - New Microsoft Word Document (2)

File Edit View Insert Format Tools Table Window Help

Normal Times New Roman 10 B I U

Error handling (1): Pose a question Clippit does not understand

Which conversational rule is violated?

What would you like to do?

When I ask a librarian, s/he sticks to the conversational scripts and rules

Search

Tips Options Close

Human conversational script: *Error handling*
Librarian: "Sorry, I don't understand the question. Could you give me a title?"
Not: "What would you like to do? Borrow a book? Pay your fee? Call a taxi? Write your own book? See more..."

What misses: Knowledge of relevance to user goals

What would you like to do?

- Format table of authorities entries
- Apply formatting to information inserted by fields
- Create a form for a Web page
- Insert fields
- Insert Word fields in a mail-merge main document

See more...

When I ask a librarian, s/he sticks to the conversational scripts and rules

Search

Tips Options Close

95

Start H:\MMI99009\Text Design Microsoft PowerPoint - [...] Microsoft Word - New...

11:21 AM

Error handling (2): Avoidance of repetition

First attempt

What would you like to do?

I don't know what you mean. Please rephrase your question.

A librarian!



Second attempt

What would you like to do?

I don't know what you mean. Please rephrase your question.

A librarian!



Third attempt

What would you like to do?

I don't know what you mean. Please rephrase your question.

A librarian!



Fourth attempt, with rephrasing

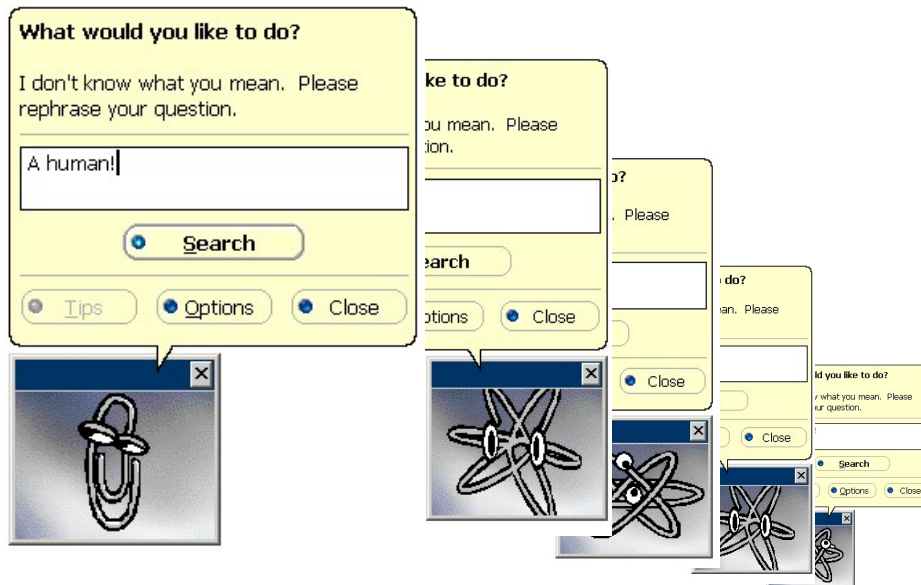
What would you like to do?

I don't know what you mean. Please rephrase your question.

A human!



What would the human conversational script suggest in this situation? Note: Are Clippit's expressions helpful to understand its communications? Are the expressions to the point?

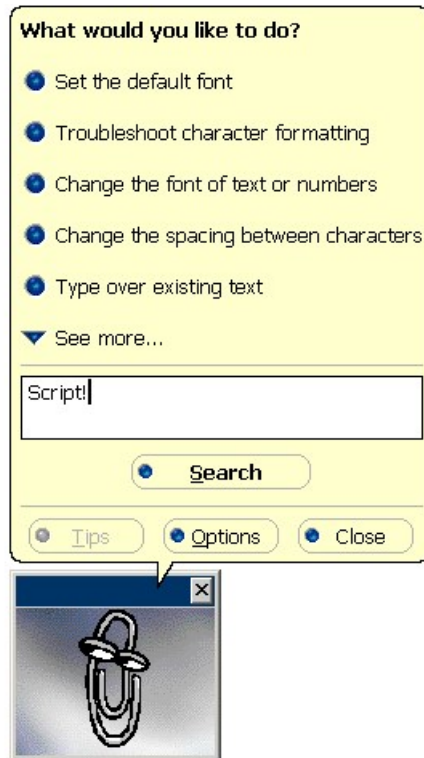


Human conversational script: *End of dialog*
 Librarian: "I am terribly sorry, but I cannot help you. I will ask my manager."

Not: "I don't know what you mean, rephrase. I don't know what you mean, rephrase, etc." (vicious loop).

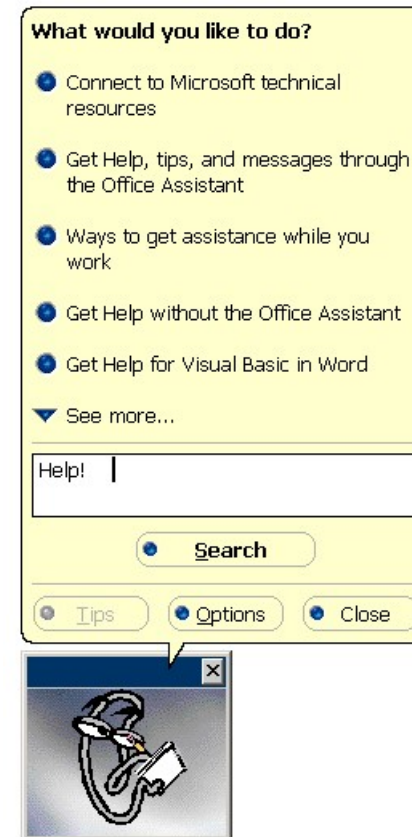
What misses: A frequency count

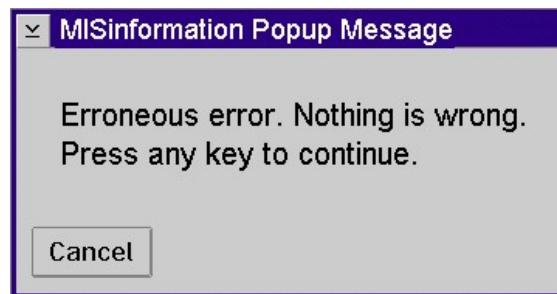
Humans interpreting machines: conclusions



Human centered text in interaction design is not dressing up a conventional tool with some anthropomorphism.

It requires deep understanding of the structure of conversation (Scripts!) and other language behaviors, with a focus on (contextual) semantics and pragmatics.





Human-generated summary



Natural language is hard! Full NL interfaces are probably still several years off yet

This is because human language is less dependent on form ('grammar') and more on content ('semantics'), which is susceptible to situation, context, pragmatics

When designing a speech or text interface, we should be aware that we do speech acts (e.g., commands are 'directive'), which ensue conversational rules (e.g., 'be relevant to user goals')

Many human-human interactions are formal (e.g., wedding formula) and make use of psychological scripts (e.g., how to open/close a meeting). These are the low hanging fruit for NLP applications

Human-generated summary



- Communicating with systems in real-world situations through natural language is much needed
- But it is also an AI-complete problem
- Generic technologies such as probabilistic reasoning, deep learning need to be integrated and adapted to deal with complex multimodal settings
- Flexibility, integrated learning, robustness for errors, uncertainties and noise, incomplete data streams
- Theory of mind, affective relations, common ground
- Negotiation of goals and social roles
- Interdisciplinary R&D is key

References

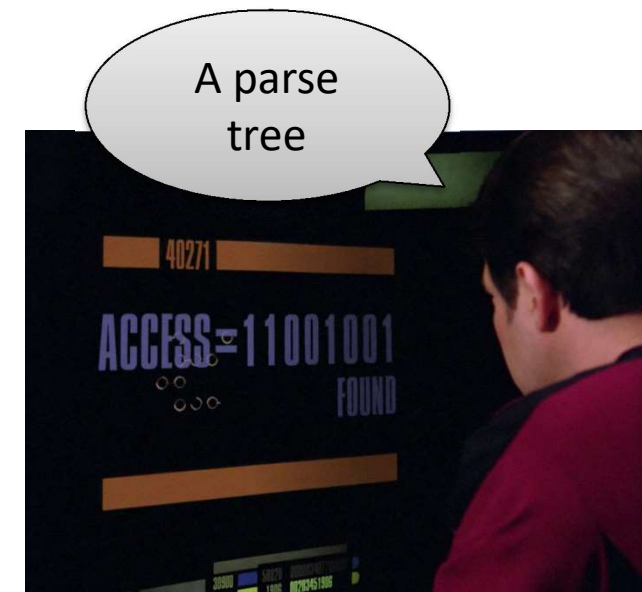
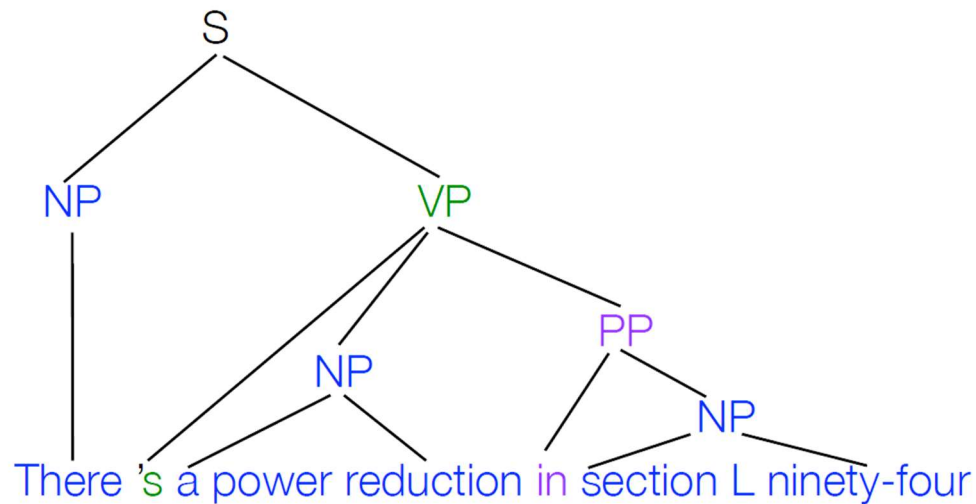
- Bahdanau, D., Cho, K., & Bengio, Y. (2014). Neural machine translation by jointly learning to align and translate. arXiv preprint arXiv:1409.0473
- Hendrycks, D., & Gimpel, K. (2016). Gaussian error linear units (Gelus). *arXiv preprint arXiv:1606.08415*
- Krügel, S., Ostermaier, A., & Uhl, M. (2023). ChatGPT's inconsistent moral advice influences users' judgment. *Scientific Reports*, 13(1), 4569.
- Mavridis, Nikolaos. "A review of verbal and non-verbal human-robot interactive communication." *Robotics and Autonomous Systems* 63 (2015): 22-35
- Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). Whose opinions do language models reflect?. *arXiv preprint arXiv:2303.17548*
- Tangiuchi, Tadahiro, Daichi Mochihashi, Takayuki Nagai, Satoru Uchida, Naoya Inoue, Ichiro Kobayashi, Tomoaki Nakamura, Yoshinobu Hagiwara, Naoto Iwahashi, and Tetsunari Inamura. "Survey on frontiers of language and robotics." *Advanced Robotics* 33, no. 15-16 (2019): 700-730.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Proceedings of Advances in Neural Information Processing Systems (NIPS'17)*, 30.
- Wikipedia (2020): Normal distribution. Wikipedia, the free encyclopedia. https://en.wikipedia.org/wiki/Normal_distribution#Cumulative_distribution_function
- Wikipedia (2020): Error function. Wikipedia, the free encyclopedia. https://en.wikipedia.org/wiki/Error_function
- Vossen, Piek, Lenka Bajčetić, Selene Baez, Suzana Bašić, and Bram Kraaijeveld. "Modelling context awareness for a situated semantic agent." In *International and Interdisciplinary Conference on Modeling and Using Context*, pp. 238-252. Springer, Cham, 2019

Exam Sample Item (Open Question)

Category “Knowledge” and “Insight”

Commander Riker uses natural language to operate his computer.

A. How is the structure called to analyze groups of related tokens that form larger constructs?



Exam Sample Item (Open Question)

Category “Knowledge” and “Insight”

Commander Riker uses natural language to operate his computer.

B. Where in the compiler architecture does the parse tree fit in and why is it exactly there?

